



UTM

UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

SEMESTER 2

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SECP1033 COMPUTER ORGANIZATION AND ARCHITECTURE

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1.0 Introduction - Problem Background, aim, objective, scopes (computer specification & benchmarking tools features)

CPU benchmarking is essential for measuring and comparing the performance of different processors. By running specific tests, we can evaluate a CPU's efficiency, speed, and overall capability. This helps users and professionals make informed decisions regarding hardware purchases and system optimizations.

Benchmarking involves running standardized tests on a CPU to measure its performance across various tasks, generating scores or metrics that reflect the CPU's abilities in different scenarios. The main objective is to provide a reliable and repeatable method for comparing CPUs under identical conditions.

There are different types of benchmarks used for this purpose. Synthetic benchmarks use artificial tasks to test specific aspects of CPU performance, such as complex calculations and data processing, to measure the CPU's maximum potential. Real-world benchmarks evaluate CPU performance using tasks that simulate actual usage scenarios like web browsing, video playback, and gaming, providing practical insights into everyday performance. Application benchmarks measure CPU performance using specific software applications, focusing on tasks like rendering graphics, processing large datasets, and running simulations. They help users understand how well a CPU performs with their regular software.

Aim

The aim of this project is to measure and compare the performance of various CPUs using different benchmarking tools and methods. By doing this, we can provide a detailed analysis of how well different CPUs perform. This will help users choose the right CPU for their needs, whether for gaming, professional work, or general use. The project will also highlight any performance issues and suggest areas for improvement.

Objectives :

1. Understand the principles of CPU benchmarking.
2. Compare the performance of different CPUs.
3. Analyze the impact of different workloads on CPU performance.
4. Learn to use free benchmarking tools and interpret their results.

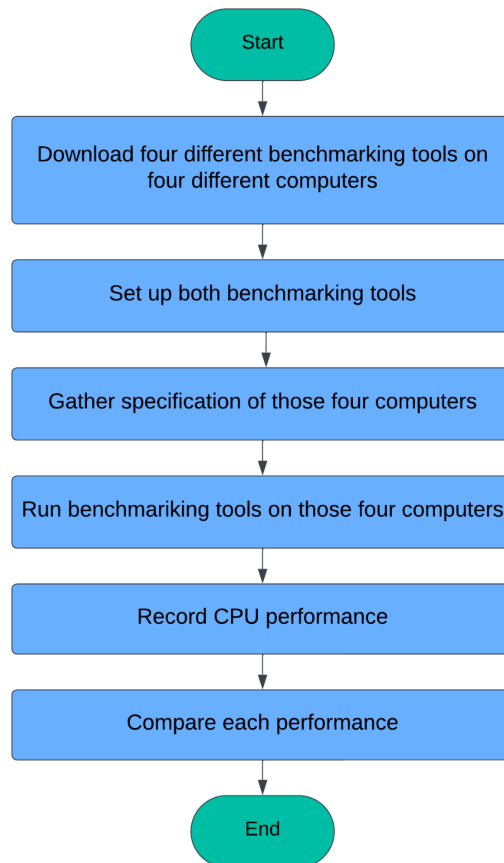
Scopes (computer specification & benchmarking tools features) :

Specifications	Computer Type			
	Lenovo 20JTS0HT00	ASUSTeK COMPUTER INC. ASUS TUF Gaming F15	Dell Inc. Latitude E5570	ASUSTeK COMPUTER INC. VivoBook_ASUS Laptop
Processor Type	Intel Core i7 6600U	Intel Core i5 10300H	Intel Core i5 6300U	Intel Celeron N4000
RAM Size	8.00 GBytes	8.00 GBytes	8.00 GBytes	8.00 GBytes
Motherboard	Lenovo 20JTS0HT00	ASUSTeK COMPUTER INC. FX06LHB	Dell Inc. 0R3VX4	ASUSTeK COMPUTER INC. X415MA
Bus Specs.	PCI-Express 3.0 (8.0 GT/s)	PCI-Express 3.0 (8.0 GT/s)	PCI-Express 3.0 (8.0 GT/s)	PCI-Express 2.0 (5.0 GT/s)

In this project, we use CPU-Z, Geekbench, Windows Task Manager and MiniTool Partition Wizard Free 12.8 - Disk Benchmark to benchmark the computers.

1. **CPU-Z:** CPU-Z is a software tool that gives detailed information about the hardware components of a computer. It focuses on the CPU but also provides data about other parts like the motherboard, memory, and system cache. It's useful for understanding your computer's specifications and monitoring real-time CPU performance.
2. **Geekbench:** Geekbench is a tool used for benchmarking CPUs across different platforms (like Windows, macOS, Linux, and mobile OS). It measures CPU and memory performance through various tests, providing scores that can be compared across different devices and configurations.
3. **Windows Task Manager:** To open the Windows Task Manager, press Ctrl + Alt + Delete and select "Task Manager" from the options. It's a built-in utility in Windows that lets you monitor the applications, processes, and performance of your computer. You can use it to check CPU usage, memory usage, disk activity, and network activity.
4. **MiniTool Partition Wizard Free 12.8 - Disk Benchmark:** MiniTool Partition Wizard Free 12.8 includes a disk benchmark feature. This tool tests the read and write speeds of your computer's storage devices (like hard drives or SSDs). It helps you assess the performance of your disk and compare it with standard benchmarks to see how well it performs.

2.0 The flowchart of the project (benchmarking methodology / procedure)



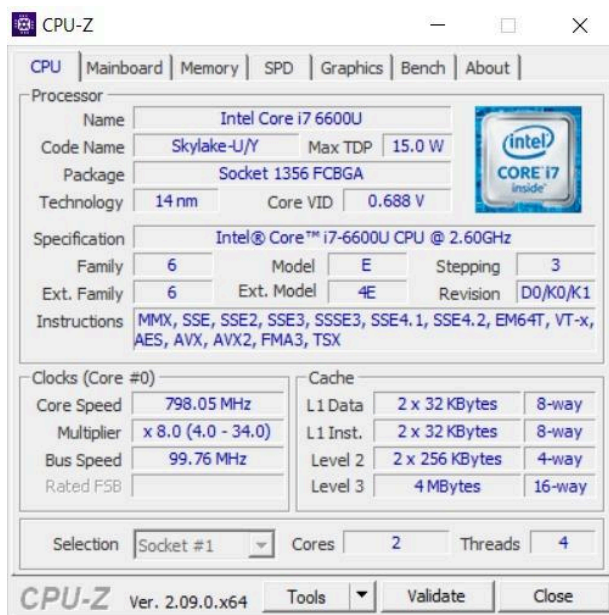
3.0 Benchmarking results

3.1 CPU-Z

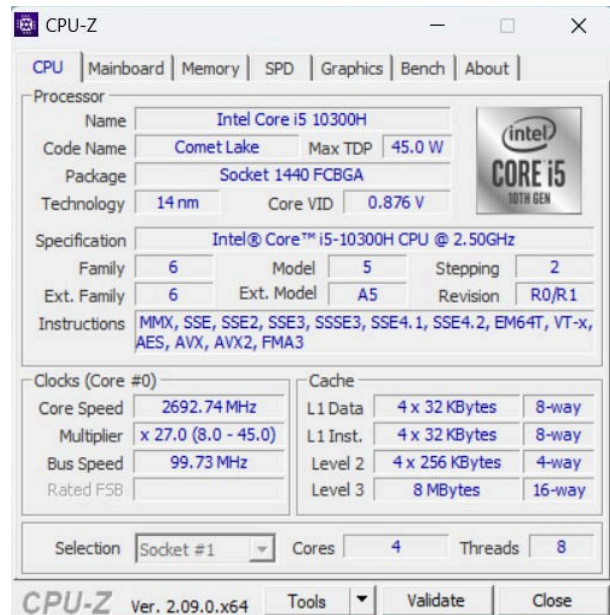
Benchmark	Computer Type			
	Lenovo 20JTS0HT00	ASUSTeK COMPUTER INC. ASUS TUF Gaming F15	Dell Inc. Latitude E5570	ASUSTeK COMPUTER INC. VivoBook_ASUSL aptop
CPU				
CPU processor (clock frequency/sp eed)	798.05 MHz	2692.72 MHz	797.66 MHz	795.64 MHz
Technology	14nm	14nm	14nm	14nm
Num of Core	2	4	2	2
Bus Speed	99.76 MHz	99.73 MHz	99.71 MHz	99.52 MHz
Cache				
L1 Data	2x32 KBytes : 8-way	4x32 KBytes : 8-way	2x32 KBytes : 8-way	2x24 KBytes : 6-way
L2 Inst	2x32 KBytes : 8-way	4x32 KBytes : 8-way	2x32 KBytes : 8-way	2x32 KBytes : 6-way
Level 2	2x256 KBytes : 4-way	4x256 KBytes : 4-way	2x256 KBytes : 4-way	4MBytes : 16-way
Level 3	4 MBytes : 12-way	8 MBytes : 16-way	3 MBytes : 12-way	-
Memory				
Type	DDR4	DDR4	DDR4	DDR4

Size	8GBytes	8GBytes	8GBytes	8GBytes
Motherboard				
Manufacturer & Model	Lenovo 20JTS0HT00	ASUSTek COMPUTER INC. FX506LHB	Dell Inc. 0R3VX4	ASUSTeK COMPUTER INC.

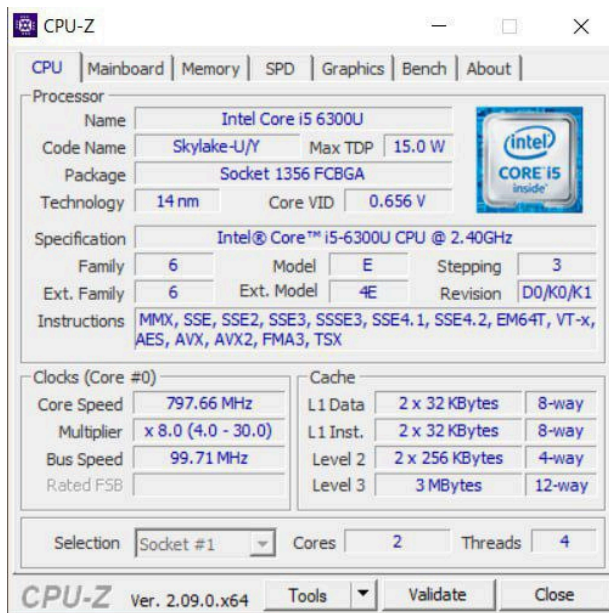
3.1.1 CPU-Z (CPU)



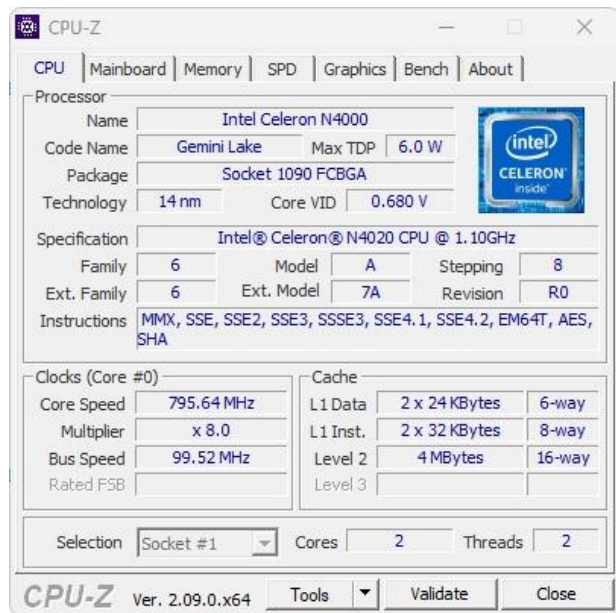
Lenovo 20JTS0HT00



ASUSTeK COMPUTER INC. ASUS TUF Gaming F15

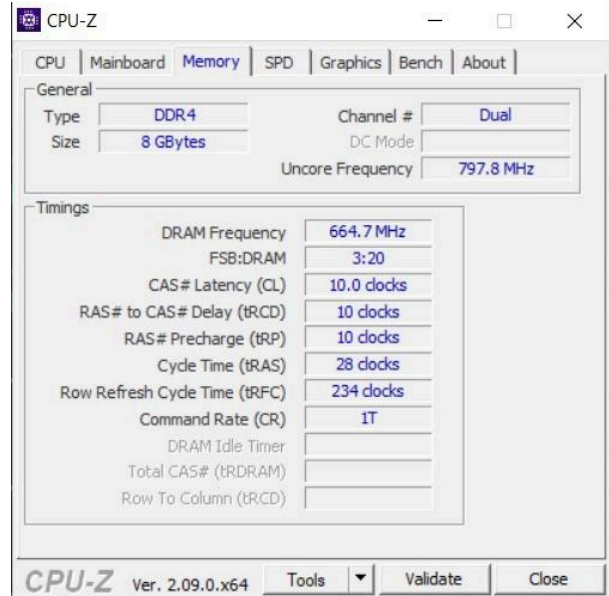


Dell Inc. Latitude E5570

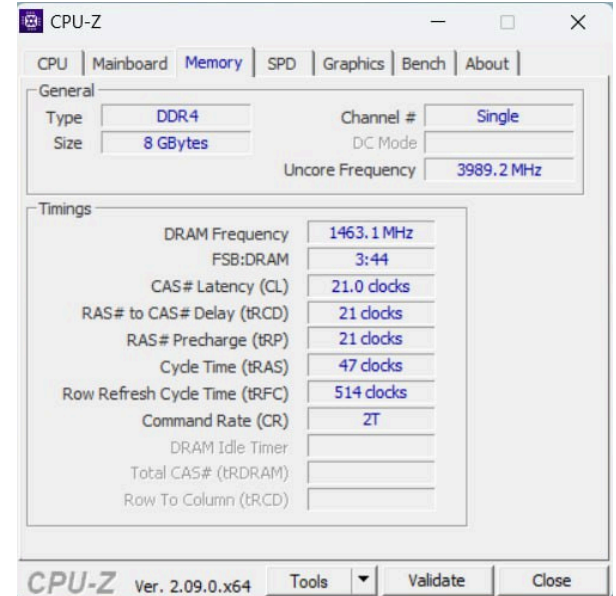


ASUSTeK COMPUTER
INC.VivoBook_ASUSLaptop

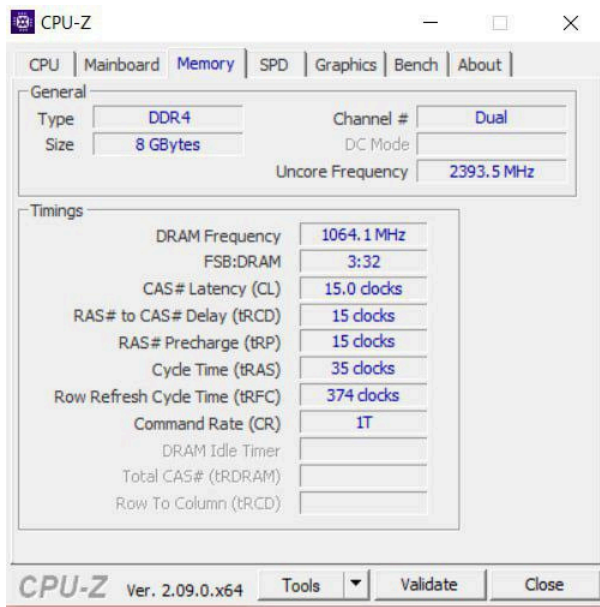
3.1.2 CPU-Z (Memory)



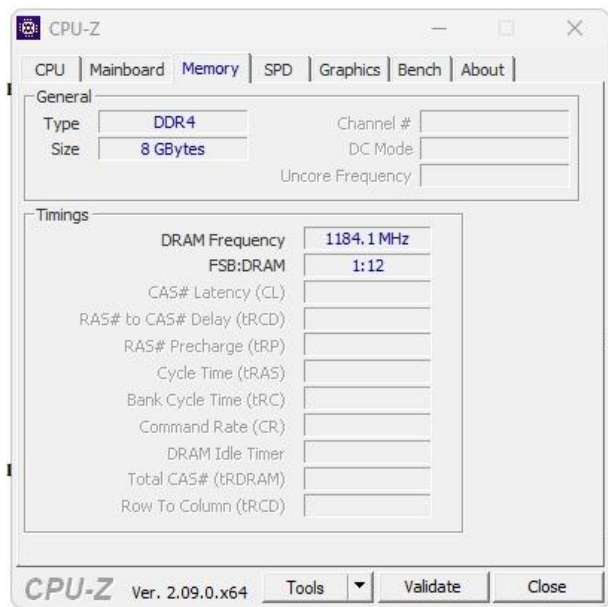
Lenovo 20JTS0HT00



ASUSTeK COMPUTER INC. ASUS TUF Gaming F15

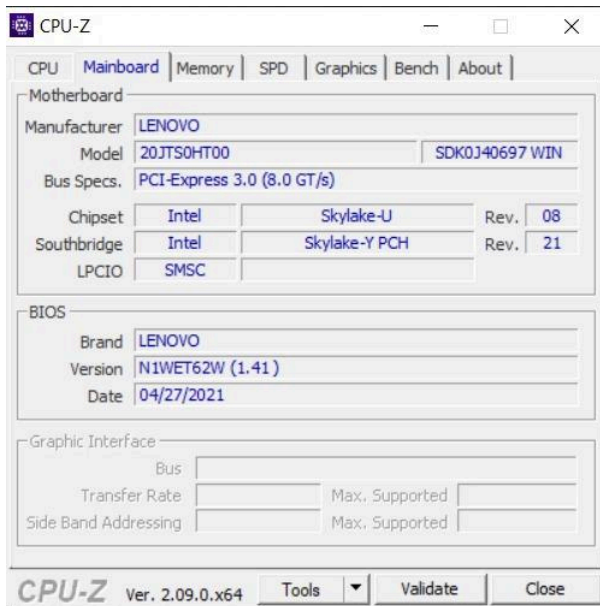


Dell Inc. Latitude E5570

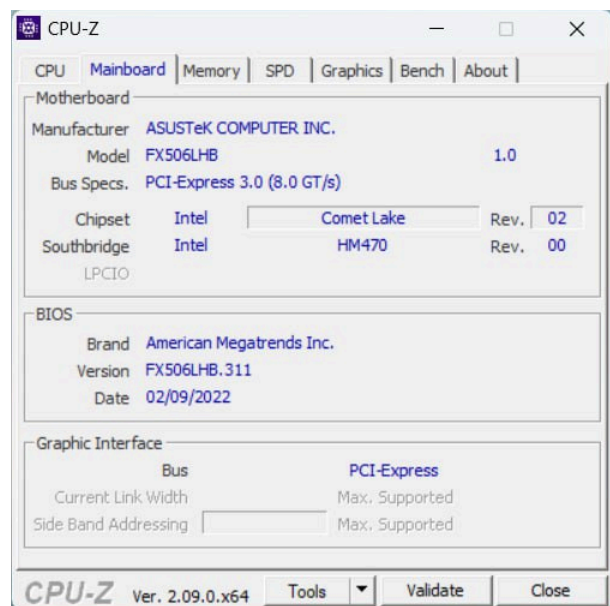


ASUSTeK COMPUTER INC.
VivoBook_ASUSLaptop

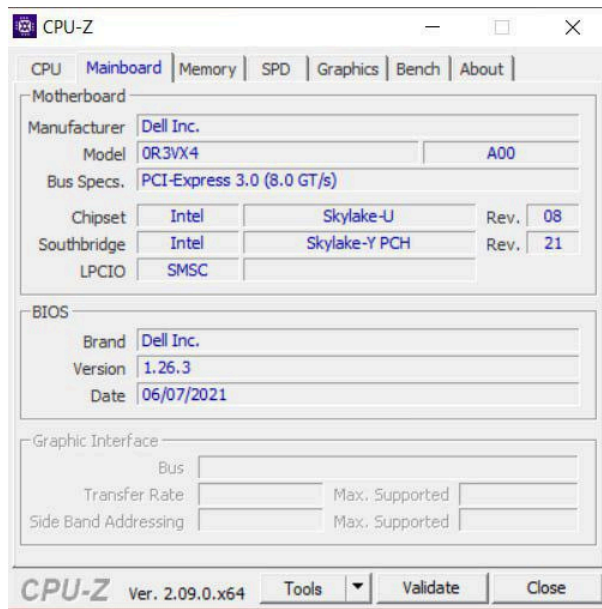
3.1.3 Motherboard



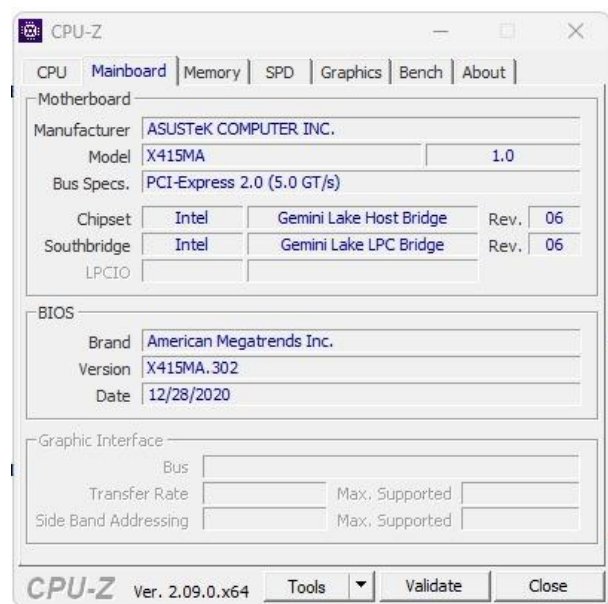
Lenovo 20JTS0HT00



ASUSTeK COMPUTER INC. ASUS TUF Gaming F15



Dell Inc. Latitude E5570



ASUSTeK COMPUTER INC.
VivoBook_ASUSLaptop

3.2 Geekbench

Benchmark	Computer Type			
	Lenovo 20JTS0HT00	ASUSTeK COMPUTER INC. ASUS TUF Gaming F15	Dell Inc. Latitude E5570	ASUSTeK COMPUTER INC. VivoBook_ASUSLa ptop
CPU				
Maximum Frequency	3391 MHz	4491 MHz	2993 MHz	2685 MHz
Cache				
LI Instruction Cache	32.0 KB x 2	32.0 KB x 4	32.0 KB x 2	32.0 KB x 2
L1 Data Cache	32.0 KB x 2	32.0 KB x 2	32.0 KB x 2	24.0 KB x 2
L2 Cache	256.0 KB x 2	256.0 KB x 4	256.0 KB x 2	4.0 MB x 1
L3 Cache	4.0 KB x 1	8.0 KB x 1	3.0 KB x 1	-
Memory				
Type	DDR4 SDRAM	DDR4 SDRAM	DDR4 SDRAM	DDR4 SDRAM
Size	7.65 GB	8.00 GB	8.00 GB	8.00 GB
Core Performance				
Single-Core Performan ce	560	1537	940	347
Multi-Core Performan ce	1400	4402	1910	559

CPU Information	
Name	Intel Core i7-6600U
Topology	1 Processor, 2 Cores, 4 Threads
Identifier	GenuineIntel Family 6 Model 78 Stepping 3
Base Frequency	2.79 GHz
Cluster 1	2 Cores
Maximum Frequency	3391 MHz
Package	Socket 1356 FCBGA
Codename	Skylake-U/Y
L1 Instruction Cache	32.0 KB x 2
L1 Data Cache	32.0 KB x 2
L2 Cache	256 KB x 2
L3 Cache	4.00 MB x 1

Memory Information	
Size	7.65 GB Task Manager
Transfer Rate	2128 MT/s
Type	DDR4 SDRAM
Channels	2

Single-Core Performance	
Single-Core Score	560
File Compression	533 76.5 MB/sec
Navigation	658 3.96 routes/sec
HTML5 Browser	620 12.7 pages/sec
PDF Renderer	633 14.6 Mpixels/sec
Photo Library	468 6.35 images/sec
Clang	605 2.98 Klines/sec
Text Processing	524 41.9 pages/sec
Asset Compression	616 19.1 MB/sec
Object Detection	251 7.52 images/sec
Background Blur	764 3.16 images/sec
Horizon Detection	887 27.6 Mpixels/sec
Object Remover	425 32.7 Mpixels/sec
HDR	590 17.3 Mpixels/sec
Photo Filter	678 6.72 images/sec
Ray Tracer	481 465.4 Kpixels/sec
Structure from Motion	687 21.7 Kpixels/sec

Multi-Core Performance	
Multi-Core Score	1400
File Compression	978 140.4 MB/sec
Navigation	1558 9.39 routes/sec
HTML5 Browser	1974 40.4 pages/sec
PDF Renderer	1229 28.3 Mpixels/sec
Photo Library	1336 18.1 images/sec
Clang	1538 7.57 Klines/sec
Text Processing	615 49.2 pages/sec
Asset Compression	2318 71.8 MB/sec
Object Detection	751 22.5 images/sec
Background Blur	1377 5.70 images/sec
Horizon Detection	1816 56.5 Mpixels/sec
Object Remover	1150 88.4 Mpixels/sec
HDR	1682 49.4 Mpixels/sec
Photo Filter	2055 20.4 images/sec
Ray Tracer	2002 1.94 Mpixels/sec
Structure from Motion	2047 64.8 Kpixels/sec

3.2.1 Lenovo 20JTS0HT00

CPU Information	
Name	Intel Core i5-10300H
Topology	1 Processor, 4 Cores, 8 Threads
Identifier	GenuineIntel Family 6 Model 165 Stepping 2
Base Frequency	2.50 GHz
Cluster 1	4 Cores
Maximum Frequency	4491 MHz
Package	Socket 1440 FCBGA
Codename	Comet Lake
L1 Instruction Cache	32.0 KB x 4
L1 Data Cache	32.0 KB x 4
L2 Cache	256 KB x 4
L3 Cache	8.00 MB x 1

Memory Information	
Size	8.00 GB
Type	DDR4 SDRAM
Channels	1

Single-Core Performance	
Single-Core Score	1537
File Compression	1461 209.8 MB/sec
Navigation	1794 10.8 routes/sec
HTML5 Browser	1580 32.3 pages/sec
PDF Renderer	1696 39.1 Mpixels/sec
Photo Library	1303 17.7 images/sec
Clang	1681 8.28 Klines/sec
Text Processing	1555 124.6 pages/sec
Asset Compression	1712 53.1 MB/sec
Object Detection	688 20.6 images/sec
Background Blur	2061 8.53 images/sec
Horizon Detection	2290 71.2 Mpixels/sec
Object Remover	1420 109.2 Mpixels/sec
HDR	1557 45.7 Mpixels/sec
Photo Filter	1766 17.5 images/sec
Ray Tracer	1330 1.29 Mpixels/sec
Structure from Motion	1773 56.1 Kpixels/sec

Multi-Core Performance	
Multi-Core Score	4402
File Compression	2092 300.4 MB/sec
Navigation	6121 36.9 routes/sec
HTML5 Browser	4059 83.1 pages/sec
PDF Renderer	5569 128.4 Mpixels/sec
Photo Library	4690 63.7 images/sec
Clang	7440 36.6 Klines/sec
Text Processing	1980 158.6 pages/sec
Asset Compression	8063 249.8 MB/sec
Object Detection	1754 52.5 images/sec
Background Blur	5988 24.8 images/sec
Horizon Detection	5968 185.7 Mpixels/sec
Object Remover	5024 386.3 Mpixels/sec
HDR	4703 138.0 Mpixels/sec
Photo Filter	3237 32.1 images/sec
Ray Tracer	7179 6.95 Mpixels/sec
Structure from Motion	5089 161.1 Kpixels/sec

3.2.2 ASUSTeK COMPUTER INC. ASUS TUF Gaming F15

CPU Information	
Name	Intel Core i5-6300U
Topology	1 Processor, 2 Cores, 4 Threads
Identifier	GenuineIntel Family 6 Model 78 Stepping 3
Base Frequency	2.49 GHz
Cluster 1	2 Cores
Maximum Frequency	2993 MHz
Package	Socket 1356 FCBGA
Codename	Skylake-U/Y
L1 Instruction Cache	32.0 KB x 2
L1 Data Cache	32.0 KB x 2
L2 Cache	256 KB x 2
L3 Cache	3.00 MB x 1

Memory Information	
Size	8.00 GB
Transfer Rate	2126 MT/s
Type	DDR4 SDRAM
Channels	2

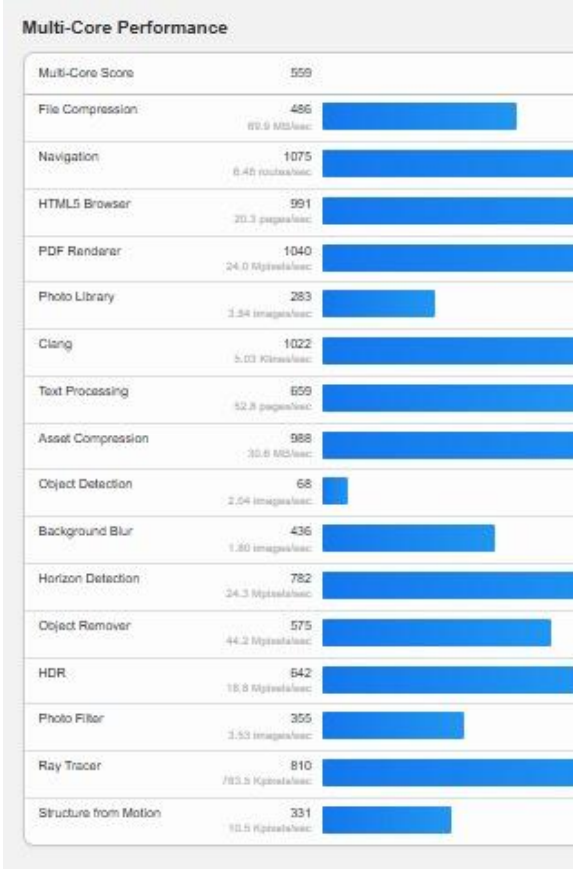
Single-Core Performance		
Single-Core Score	940	
File Compression	910 130.7 MB/sec	
Navigation	1073 6.47 routes/sec	
HTML5 Browser	1111 22.8 pages/sec	
PDF Renderer	1073 24.7 Mpixels/sec	
Photo Library	812 11.0 images/sec	
Clang	977 4.81 Klines/sec	
Text Processing	899 72.0 pages/sec	
Asset Compression	963 29.8 MB/sec	
Object Detection	432 12.9 images/sec	
Background Blur	1364 5.65 images/sec	
Horizon Detection	1393 43.4 Mpixels/sec	
Object Remover	687 52.9 Mpixels/sec	
HDR	869 25.5 Mpixels/sec	
Photo Filter	1294 12.8 images/sec	
Ray Tracer	826 799.7 Kpixels/sec	
Structure from Motion	1104 35.0 Kpixels/sec	

Multi-Core Performance		
Multi-Core Score	1910	
File Compression	1208 173.5 MB/sec	
Navigation	2423 14.6 routes/sec	
HTML5 Browser	2090 42.8 pages/sec	
PDF Renderer	2427 56.0 Mpixels/sec	
Photo Library	1810 24.6 images/sec	
Clang	2265 11.2 Klines/sec	
Text Processing	1049 84.0 pages/sec	
Asset Compression	2530 78.4 MB/sec	
Object Detection	867 25.9 images/sec	
Background Blur	2546 10.5 images/sec	
Horizon Detection	3097 96.4 Mpixels/sec	
Object Remover	1417 109.0 Mpixels/sec	
HDR	2129 62.5 Mpixels/sec	
Photo Filter	2592 25.7 images/sec	
Ray Tracer	2243 2.17 Mpixels/sec	
Structure from Motion	2390 75.7 Kpixels/sec	

3.2.3 Dell Inc. Latitude E5570

CPU Information	
Name	Intel Celeron N4020
Topology	1 Processor, 2 Cores
Identifier	GenuineIntel Family 6 Model 122 Stepping 8
Base Frequency	1.10 GHz
Cluster 1	2 Cores
Maximum Frequency	2685 MHz
Package	Socket 1090 FCBGA
Codename	Gemini Lake
L1 Instruction Cache	32.0 KB x 2
L1 Data Cache	24.0 KB x 2
L2 Cache	4.00 MB x 1

Memory Information	
Size	8.00 GB
Transfer Rate	2386 MT/s
Type	DDR4 SDRAM



3.2.4 ASUSTeK COMPUTER INC. VivoBook_ASUSLaptop

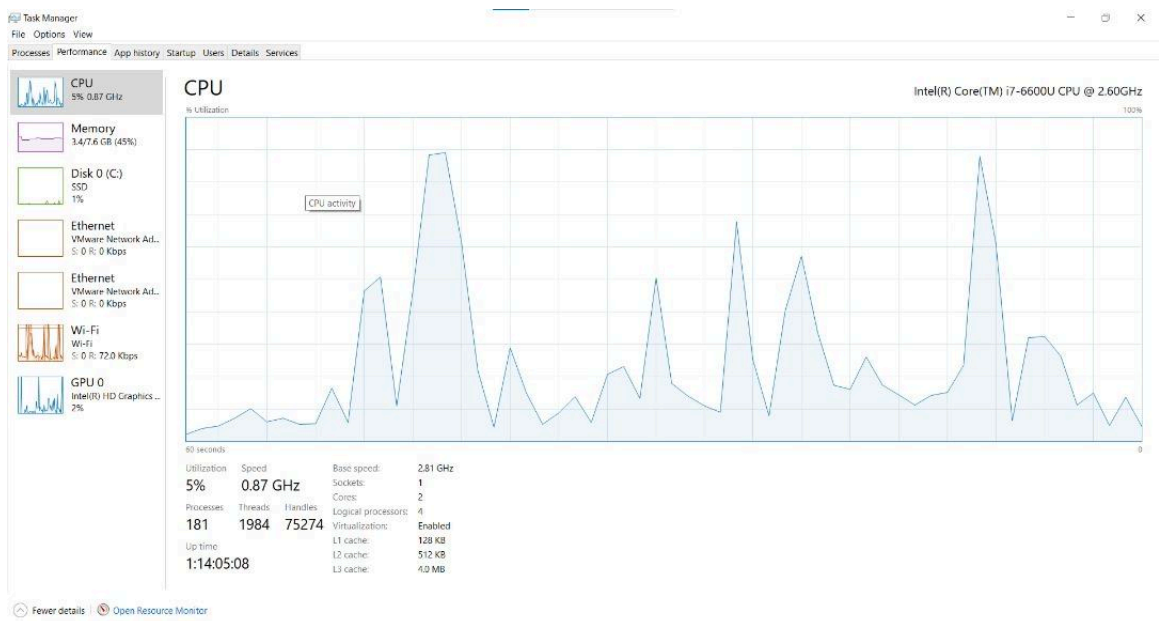
3.3 Task Manager

3.3.1 Monitor Process

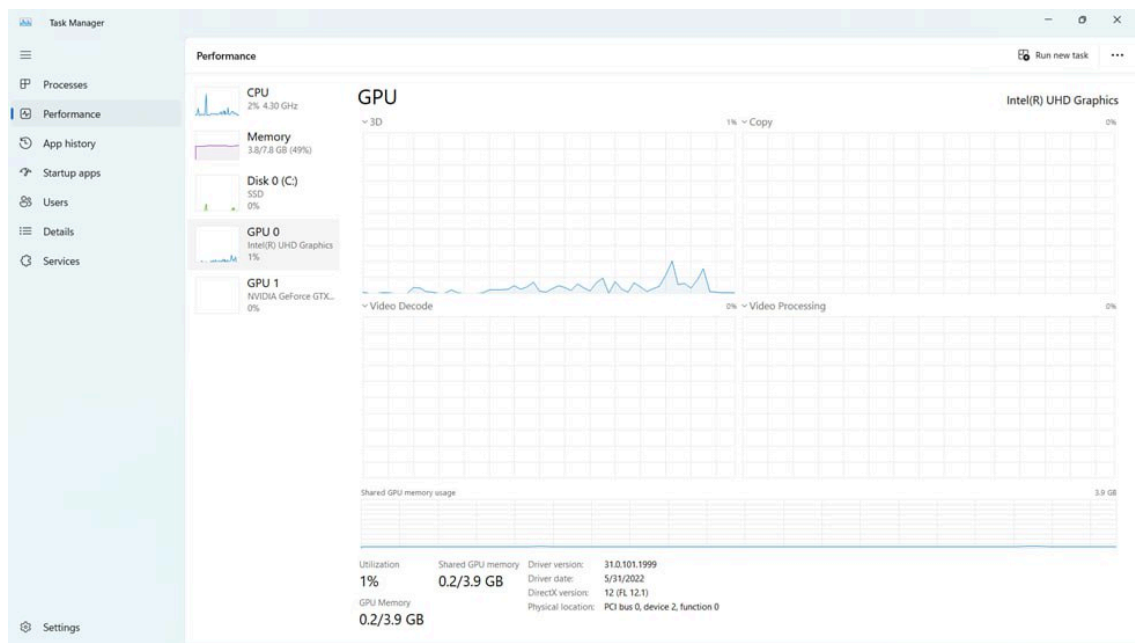
Benchmark	Computer Type			
	Lenovo 20JTS0HT00	ASUSTeK COMPUTER INC. ASUS TUF Gaming F15	Dell Inc. Latitude E5570	ASUSTeK COMPUTER INC. VivoBook_ASUS Laptop
Monitor process				
Low Load (30% - 50%)				
Number of Application Open	2	2	2	2
Memory Percentage	44%	44%	47%	48%
Middle Load (50% - 60%)				
Number of Application Open	5	5	3	3
Memory Percentage	56%	54%	53%	55%
High Load (>70%)				
Number of Application Open	13	10	6	9
Memory Percentage	80%	84%	72%	77%

3.3.2 Monitor Performance

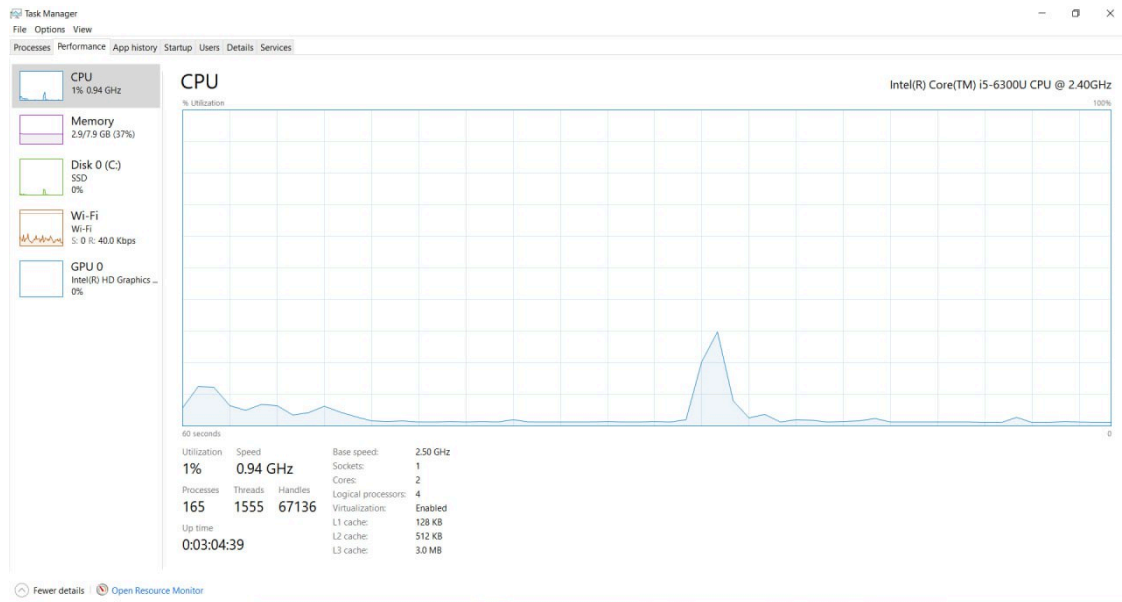
Low load



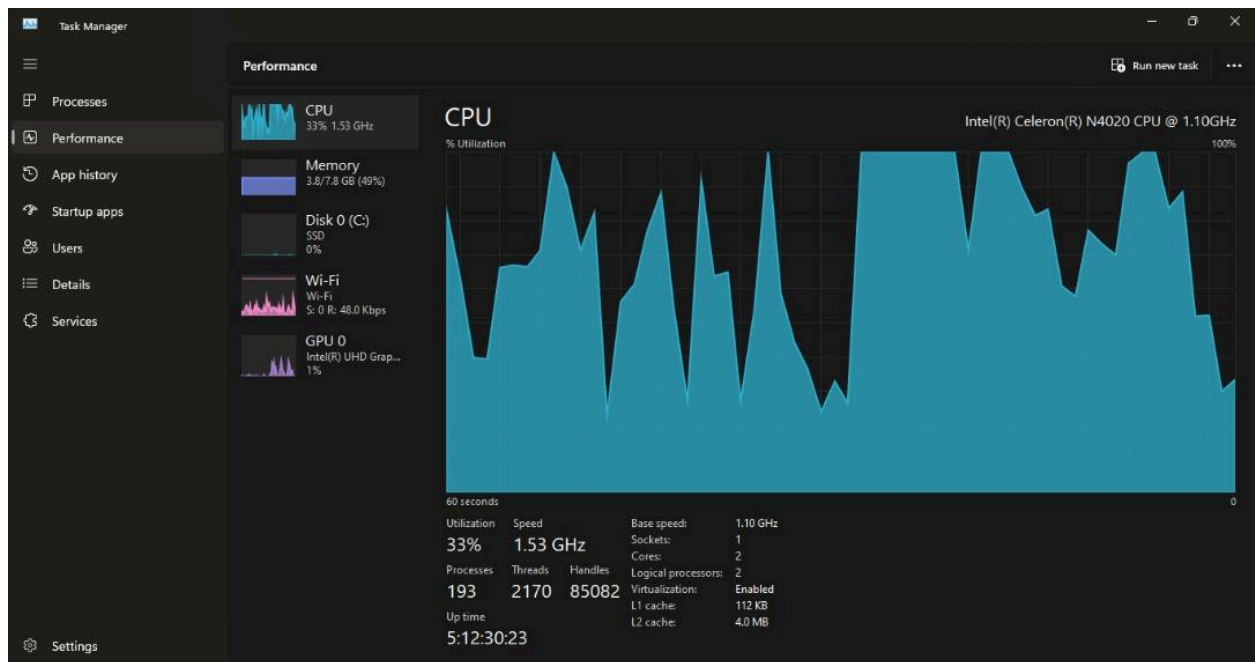
Lenovo 20JTS0HT00



ASUSTeK COMPUTER INC. ASUS TUF Gaming F15

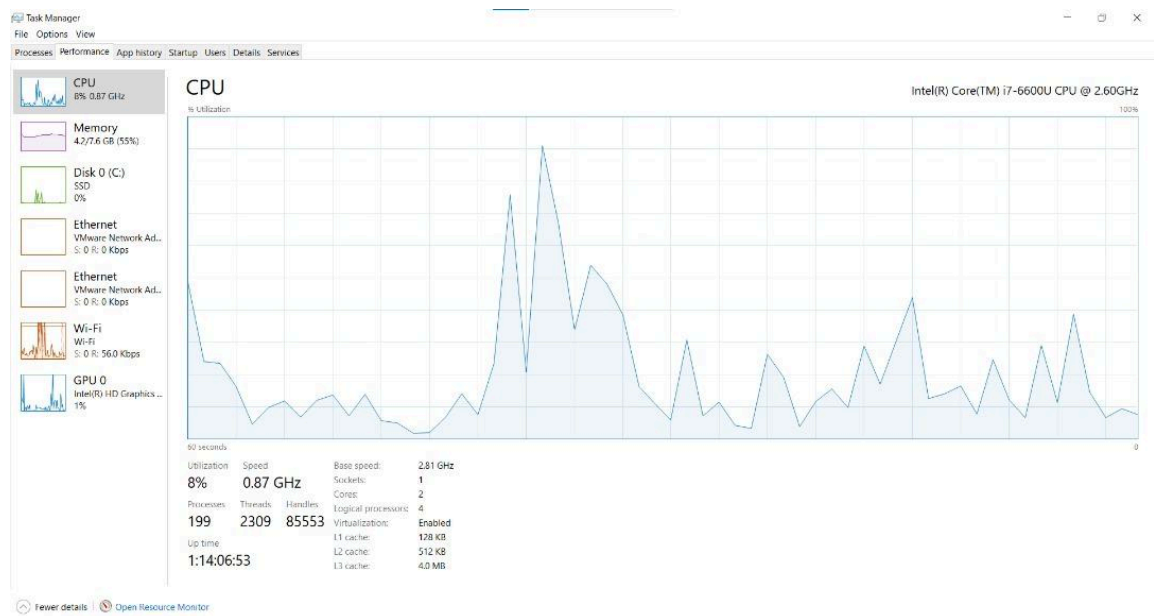


Dell Inc. Latitude E5570

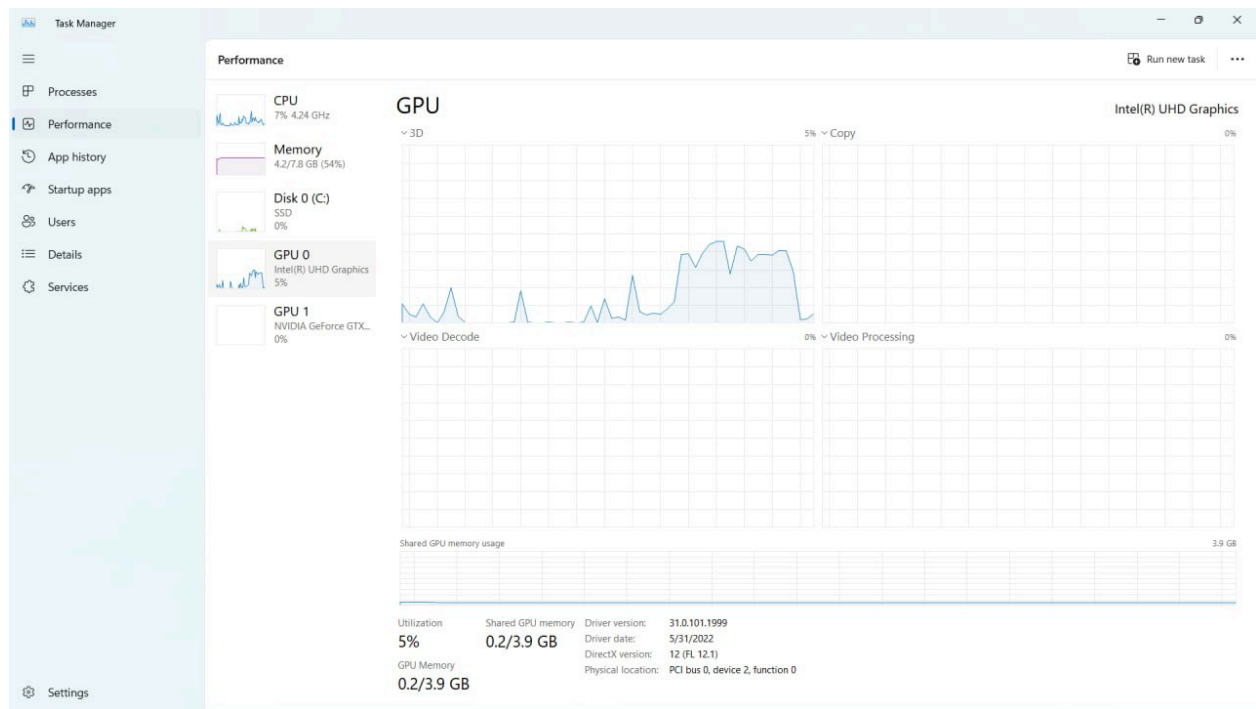


ASUSTeK COMPUTER INC. VivoBook_ASUSLaptop

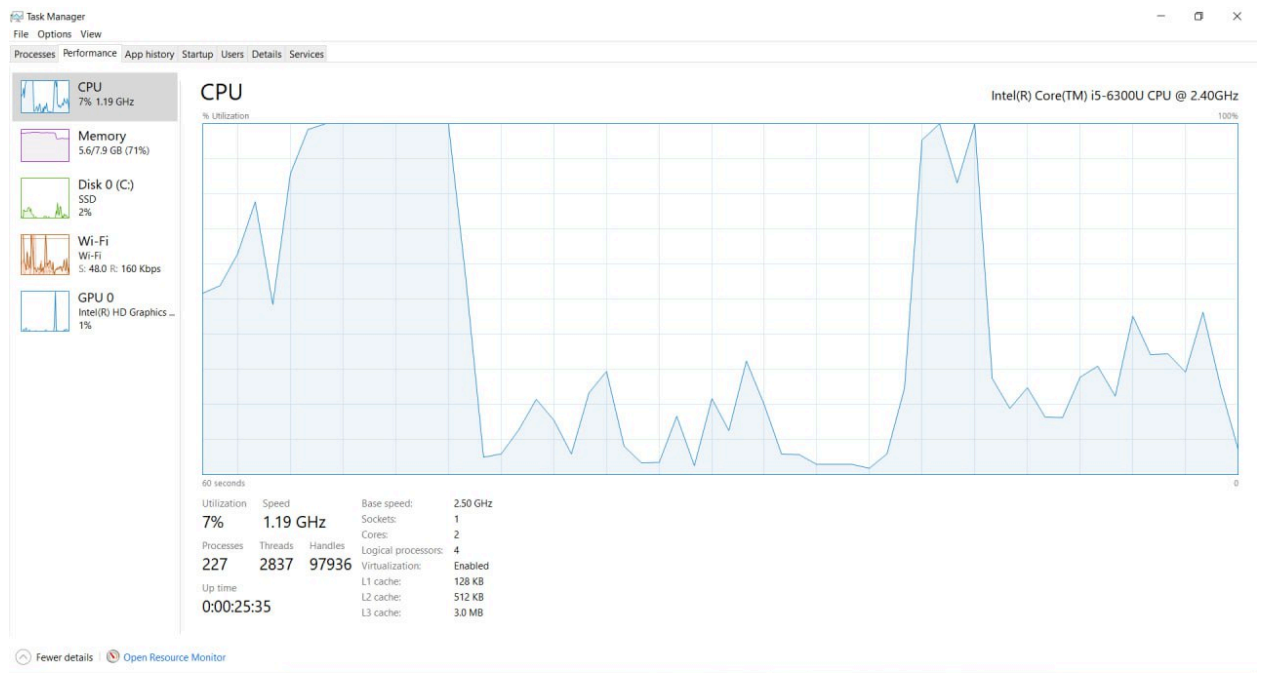
Middle Load



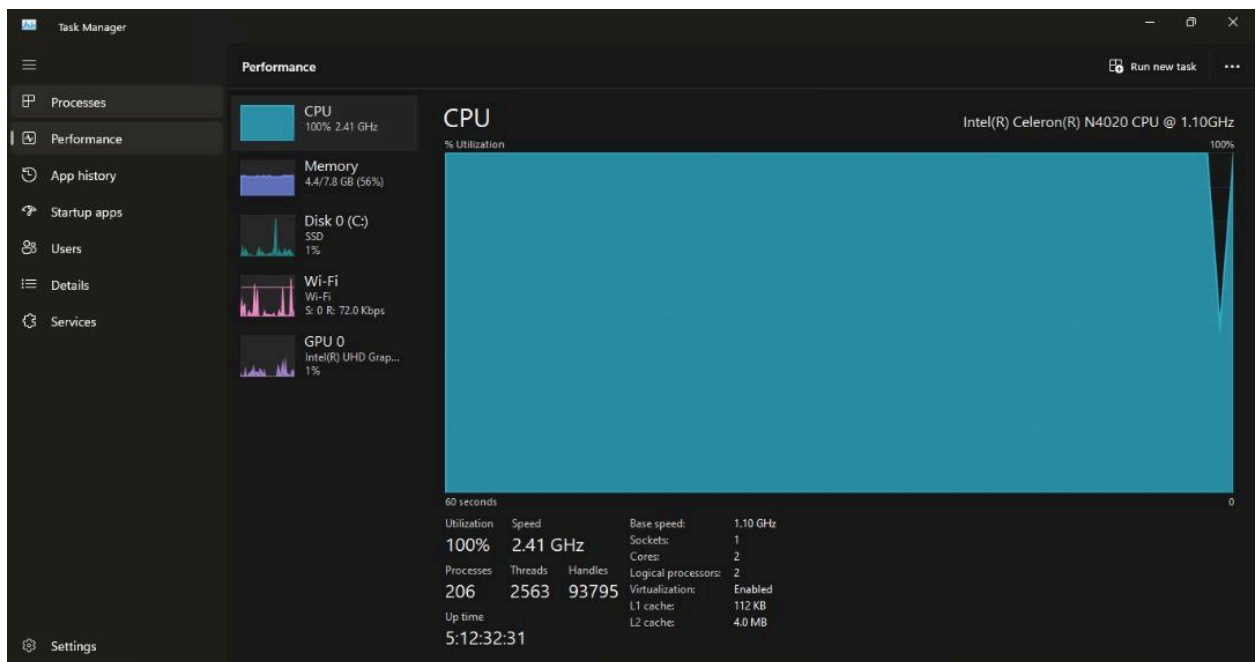
Lenovo 20JTS0HT00



ASUSTeK COMPUTER INC. ASUS TUF Gaming F15

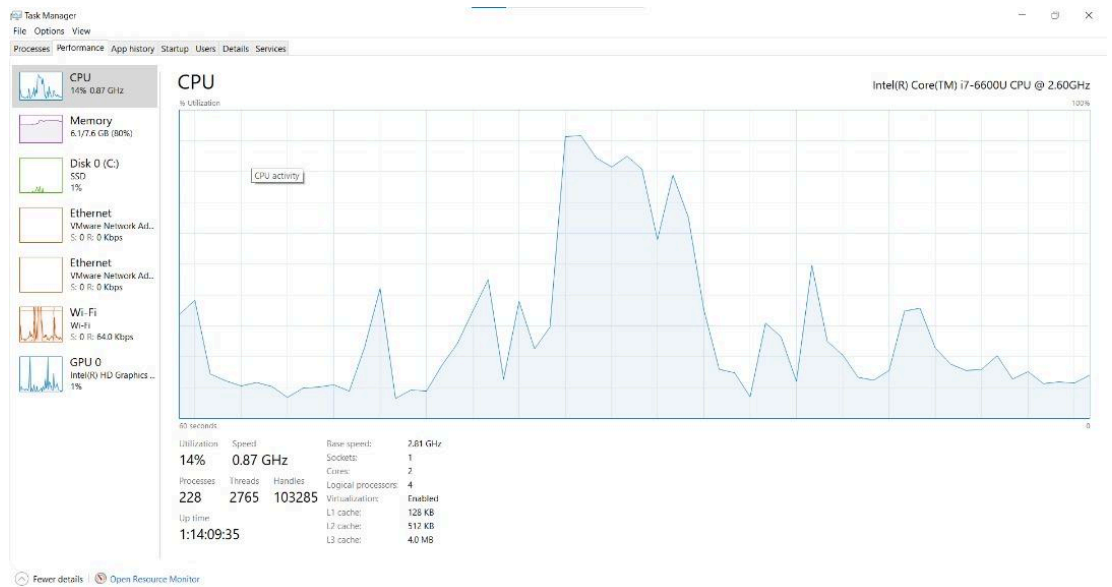


Dell Inc. Latitude E5570

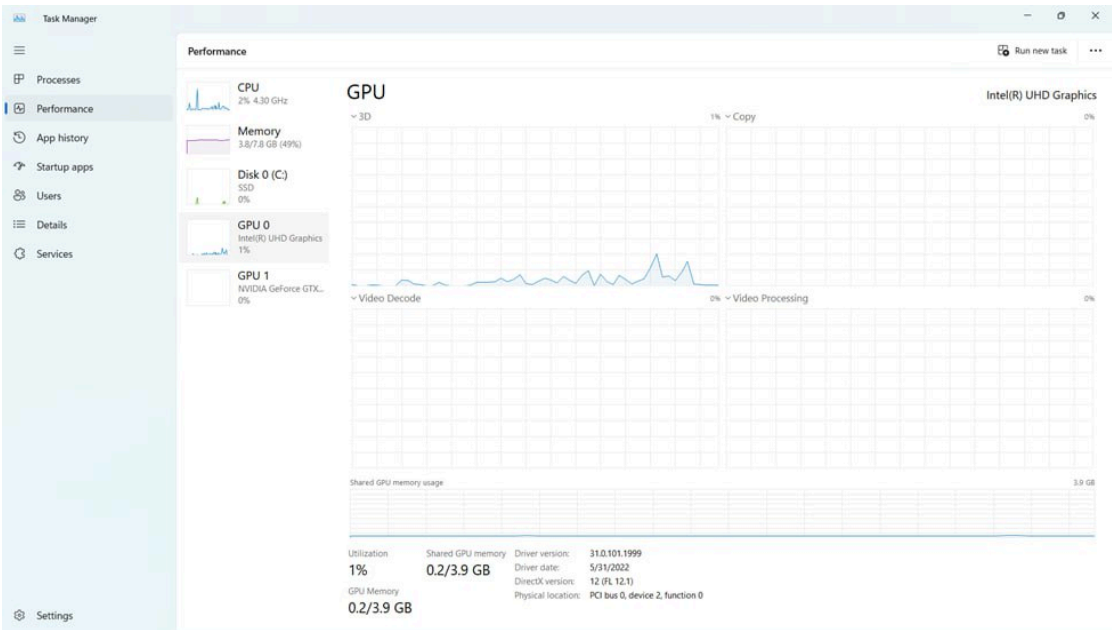


ASUSTeK COMPUTER INC. VivoBook_ASUSLaptop

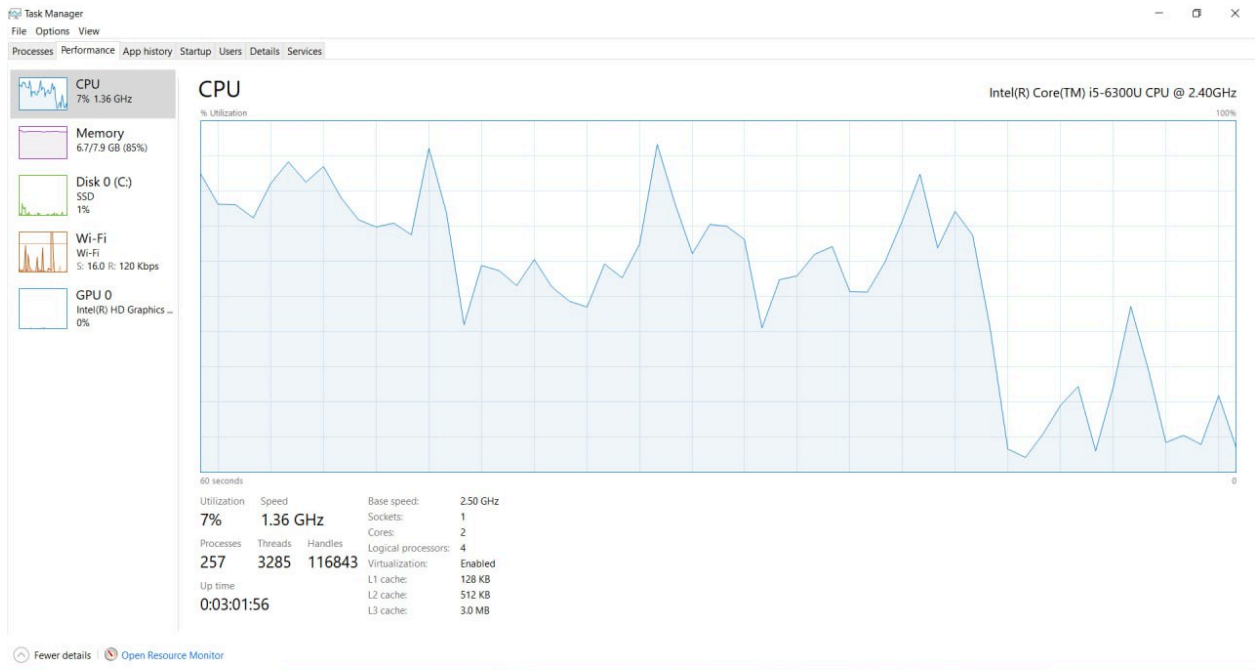
High Load



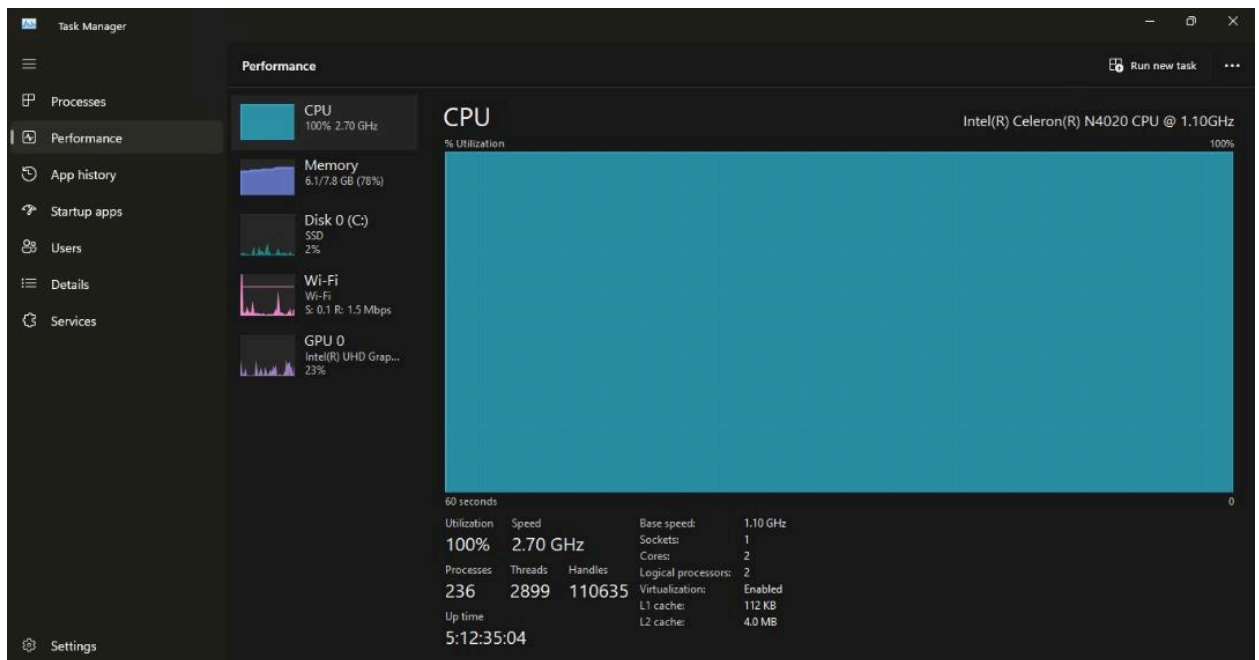
Lenovo 20JTS0HT00



ASUSTeK COMPUTER INC. ASUS TUF Gaming F15



Dell Inc. Latitude E5570



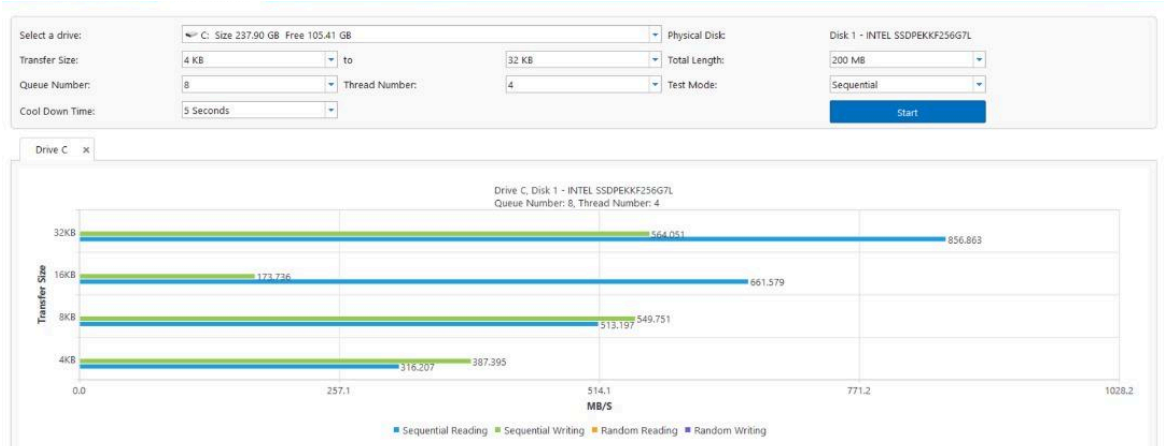
ASUSTeK COMPUTER INC. VivoBook_ASUSLaptop

3.4 MiniTool Partition Wizard

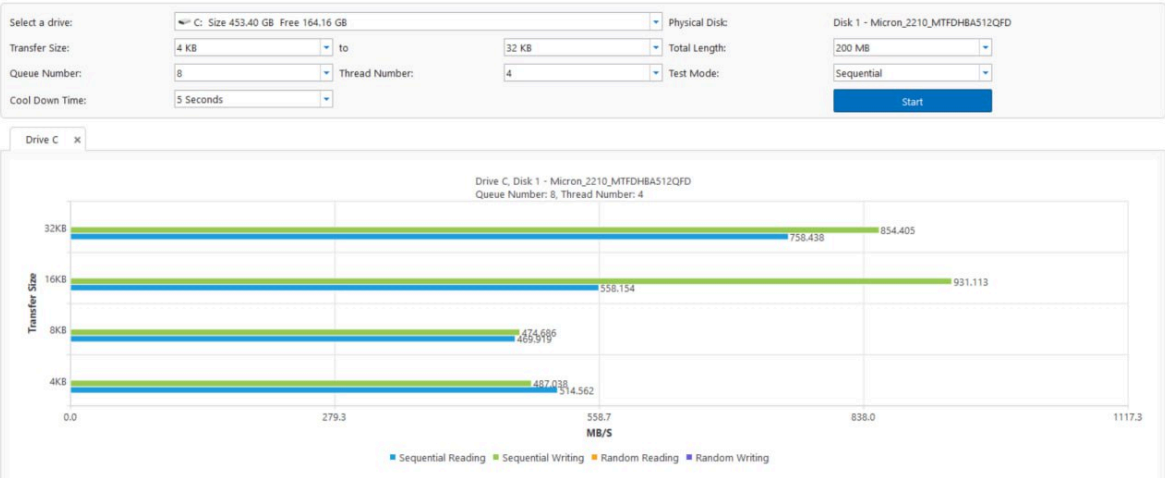
Benchmark	Computer Type			
	Lenovo 20JTS0HT00	ASUSTeK COMPUTER INC. ASUS TUF Gaming F15	Dell Inc. Latitude E5570	ASUSTeK COMPUTER INC. VivoBook_ASUS Laptop
Physical Disk	INTEL SSDPEKKF256 G7L	Micron_2210_M TFDHBA512QF D	SAMSUNG MZALQ128HC HQ-00BL2 RAID	Micron_2450_M TFDKBA256TF K
Stress Test				
Sequential Reading (MB/s)				
4KB	316.207	514.562	645.306	203.466
8KB	513.197	469.919	793.625	438.057
16KB	661.579	558.154	702.428	901.678
32KB	856.863	758.438	701.889	1547.510
Sequential Writing (MB/s)				
4KB	387.395	487.038	495.228	162.688
8KB	549.751	474.686	862.305	319.572
16KB	173.736	931.113	1130.000	615.514
32KB	564.051	854.405	1196.51	1105.870
Random Reading (MB/s)				
4KB	275.117	520.035	650.268	207.707
8KB	263.047	380.862	878.987	447.591
16KB	418.59	429.806	926.329	880.897
32KB	672.36	648.068	832.38	1095.710
Random Writing (MB/s)				

4KB	433.766	849.653	496.521	165.906
8KB	552.33	1001.59	882.522	322.460
16KB	548.455	833.701	935.474	610.020
32KB	587.691	952.713	1061.21	1047.040

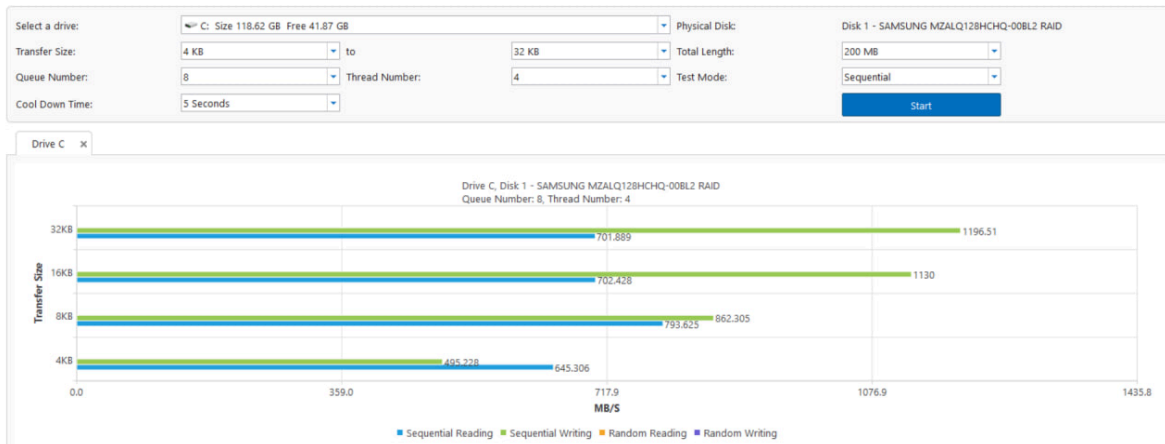
3.4.1 Test Mode: Sequential



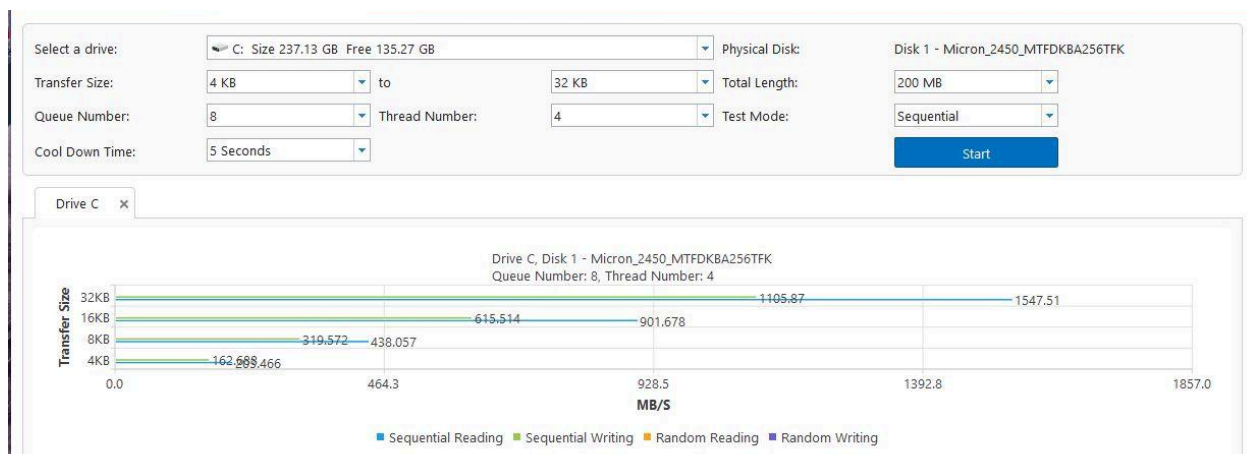
Lenovo 20JTS0HT00



ASUSTeK COMPUTER INC. ASUS TUF Gaming F15

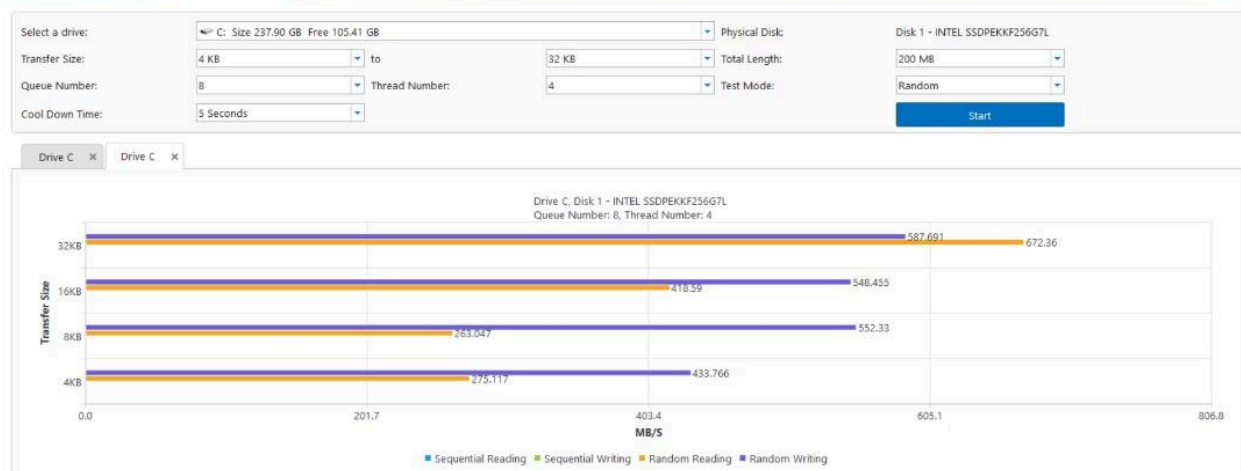


Dell Inc. Latitude E5570

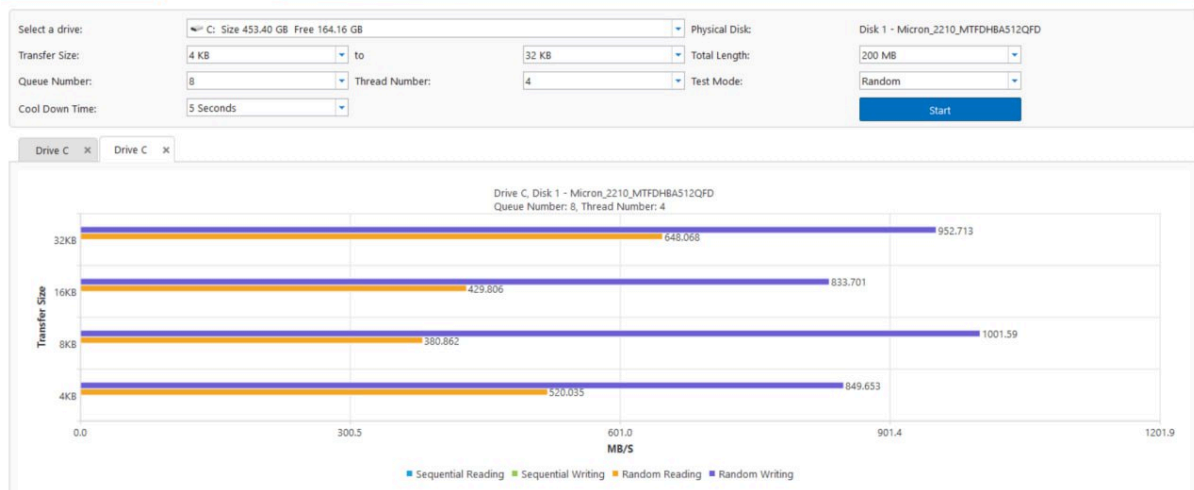


ASUSTeK COMPUTER INC. VivoBook_ASUSLaptop

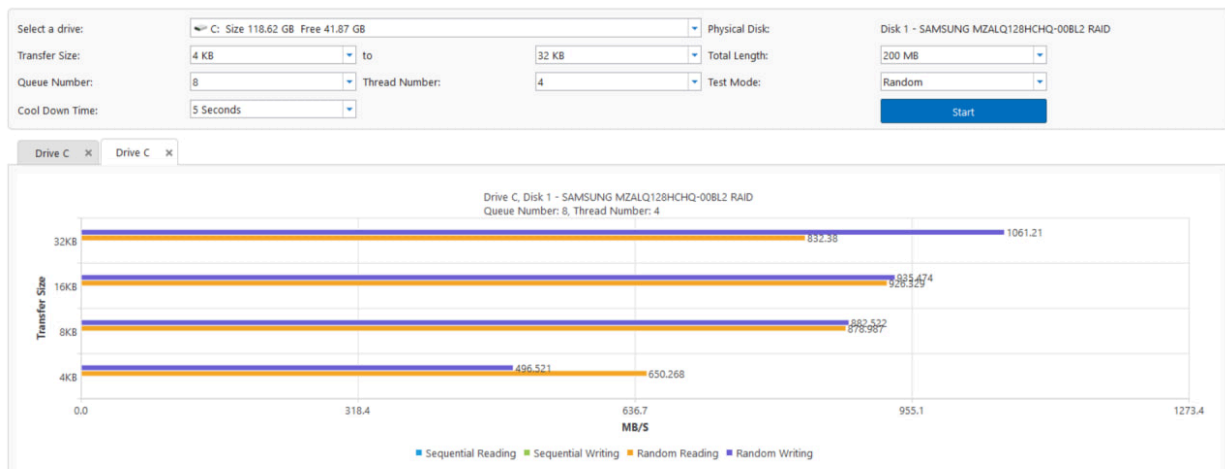
3.4.2 Test Mode: Random



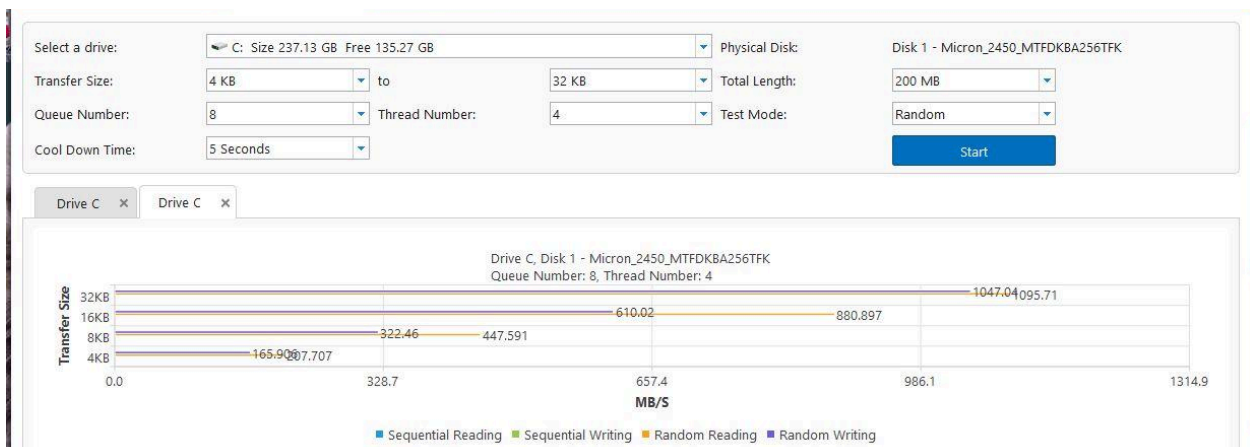
Lenovo 20JTS0HT00



ASUSTeK COMPUTER INC. ASUS TUF Gaming F15



Dell Inc. Latitude E5570



ASUSTeK COMPUTER INC. VivoBook_ASUSLaptop

4.0 Analysis, comparison, discussion

CPU-Z

1. CPU Processor (clock frequency / speed)

A high clock speed is indicative of a stable CPU that can run demanding applications and multitask efficiently. The ASUSTek COMPUTER INC. ASUS TUF Gaming F15 showed a significantly higher clock speed (2692.72 MHz) compared to Lenovo 20JTS0HT00, Dell Inc. Latitude and ASUSTeK COMPUTER INC. VivoBook_ASUSLaptop, which had speeds of 798.05 MHz, 797.66 MHz and 795.64 MHz respectively. This indicates a clear performance advantage for the ASUS TUF Gaming F15 in terms of processing power. One of the most important factors in determining how well a laptop performs is its processing speed, particularly when it comes to demanding jobs like software development, gaming and video editing. The ASUSTek COMPUTER INC. ASUS TUF Gaming F15, with its high clock speed, is well-suited for these demanding applications, offering smooth and efficient performance. The lower speeds of the other laptops like Lenovo 20JTS0HT00, Dell Inc. Latitude and ASUSTeK COMPUTER INC. VivoBook_ASUSLaptop are more appropriate for less intensive tasks such as web browsing, document editing, and media consumption.

2. CPU Technology

CPU technology is the manufacturing process used to make the Central Processing Unit (CPU). Every computer may use smaller nanometer operations, which are typically the key to higher performance and increased efficiency. All four of the PCs in our comparison use the same size of CPU technology which is 14nm. The nm measurement displays the transistor sizes within the CPU. Smaller is preferable since it allows for more transistors to fit on a given sized device, potentially meaning that it can handle modern applications more efficiently. These features increased overall system efficiency and improved multi-core application performance.

3. Number of Cores

Based on the specifications, it is clear that only ASUSTek COMPUTER INC. ASUS TUF Gaming F15 features a more powerful CPU with 4 cores while Lenovo 20JTS0HT00, Dell Inc. Latitude and ASUSTeK COMPUTER INC. VivoBook_ASUSLaptop has all features with only 2 cores. A CPU's capacity to manage multiple tasks is strongly impacted by the number of cores it has. Higher kernel speeds often translate into more processing power, which allows for quicker task execution and better overall performance. The higher number of cores are perfect for customers who utilize software that capitalizes on several cores, such scientific simulations, and video editing programmes, or who want to run multiple apps simultaneously. However, a smaller number of cores might be sufficient for everyday use.

4. Bus Speed

According to the specifications, the Lenovo 20JTS0HT00 showed a higher bus speed which is 99.76 MHz while ASUSTek COMPUTER INC. ASUS TUF Gaming F15 is 99.73 MHz, Dell Inc. Latitude is 99.71 MHz and ASUSTeK COMPUTER INC. VivoBook_ASUSLaptop is 99.52 MHz. Bus speed, which affects the data transmission rate between the CPU and other components, improves the system's overall performance. In order to process data more quickly and with less delay, a greater bus speed facilitates faster connection between the CPU, memory, and other peripherals. Furthermore, faster bus speeds enhance the effectiveness of data transport, resulting in quicker processing and more seamless functioning. On the other hand, the performance may not be significantly impacted by the little bus speed differences between each laptop. Higher bus speeds may enhance system responsiveness and data transfer rates, although real-world performance differences between 0.19 MHz and 0.24 MHz are unlikely to be valuable. Since all of the PCs in our comparison have the same range of bus speed, this indicates that all of it are good in performance of data transmission.

5. Caches

Upon examining the specifications, caches for each computer are almost the same except ASUSTeK COMPUTER INC. VivoBook_ASUSLaptop did not have a Level 3 cache. While comparing every laptop caches, we noticed that ASUSTek COMPUTER INC. ASUS TUF Gaming F15 has larger and faster cache for each listed cache in the benchmarking results, which helps to lower delays and speed up data access. Furthermore, a CPU's performance is greatly impacted by the size and speed of its cache. Caches hold frequently accessed data nearer to the CPU to reduce the amount of time it takes to retrieve it from main memory. Additionally, larger and quicker caches can store more data closer to the CPU, which reduces the amount of time it takes to get data from main memory. This is essential for tasks requiring a lot of performance and frequent access to data. When cache utilization is optimized, especially for repetitive data access patterns, application performance can be greatly increased.

6. Memory

Based on the specifications, Lenovo 20JTS0HT00, ASUSTek COMPUTER INC. ASUS TUF Gaming F15 is 99.73 MHz, Dell Inc. Latitude is 99.71 MHz and ASUSTeK COMPUTER INC. VivoBook_ASUSLaptop all have the same type of memory which is DDR4. This result means that larger and quicker memory setup adds to the overall enhanced performance. Additionally, faster and larger memory is essential for today's applications as it enables greater multitasking and massive dataset management. Faster memory rates also have an impact on improved system responsiveness, which reduces wait times and increases efficiency.

7. Motherboard

The motherboard's role is crucial as it interconnects all components of the laptop. According to the specifications, Lenovo 20JTS0HT00 used a motherboard of Lenovo 20JTS0HT00, ASUSTek COMPUTER INC. ASUS TUF Gaming F15 used ASUSTek COMPUTER INC. FX506LHB, Dell Inc. Latitude used Dell Inc. 0R3VX4 and ASUSTeK COMPUTER INC. VivoBook_ASUSLaptop used ASUSTeK COMPUTER INC.. Since the laptops used for benchmarking are all different, this results on different types of motherboard. A powerful motherboard provides steady operation to support advanced components. Moreover, more stability, versatility and speedy data transfer are offered with a better motherboard. An advanced motherboard is also expected to support modern features like quicker RAM, more storage options, and better connections.

Geekbench

When comparing CPU performance using Geekbench, we see clear differences among the computers tested. The ASUS TUF Gaming F15 has the highest maximum CPU frequency at 4491 MHz, making it the fastest. The Lenovo 20JTS0HT00 follows with 3391 MHz, then the Dell Latitude E5570 at 2993 MHz, and finally the ASUS VivoBook at 2685 MHz. A higher frequency means the CPU can handle more demanding tasks better.

Looking at the cache sizes, which help the CPU access data quickly, the ASUS TUF Gaming F15 is again the leader. It has larger L1 and L2 caches compared to the others. The L1 Instruction Cache is 32.0 KB x 4, while Lenovo and Dell have 32.0 KB x 2, and ASUS VivoBook has 24.0 KB x 2. For the L2 Cache, the ASUS TUF Gaming F15 has 256.0 KB x 4, while Lenovo and Dell have 256.0 KB x 2, and ASUS VivoBook has a single 4.0 MB cache. A bigger cache means the CPU can work faster because it can store and access more data quickly.

All four computers use DDR4 SDRAM, which is a type of memory. Their sizes are similar: Lenovo has 7.65 GB, and the others have 8.00 GB. This means they can handle memory-intensive tasks similarly well.

Geekbench also measures single-core and multi-core performance. The ASUS TUF Gaming F15 scores the highest in both 1537 for single-core and 4402 for multi-core. This means it performs best for both simple tasks that use one core and complex tasks that use multiple cores. Lenovo scores 560 and 1400, Dell scores 940 and 1910, and ASUS VivoBook scores 347 and 559. Clearly, the ASUS TUF Gaming F15 is the most powerful of the group.

Task Manager

The Task Manager benchmarks show how well these systems handle different workloads. Under low load (30% - 50%), all systems opened two applications, but their memory usage varied slightly: Lenovo and ASUS TUF Gaming F15 used 44%, Dell used 47%, and ASUS VivoBook used 48%.

Under a middle load (50% - 60%), Lenovo and ASUS TUF Gaming F15 opened five applications each, using 56% and 54% of their memory, respectively. Dell and ASUS VivoBook opened three applications each, with memory usage at 53% and 55%.

When the load is high (>70%), the ASUS TUF Gaming F15 and Lenovo show their strength. The ASUS TUF Gaming F15 opened 10 applications, using 84% of its memory, and Lenovo opened 13 applications, using 80% of its memory. Dell and ASUS VivoBook opened fewer applications (6 and 9), with memory usage at 72% and 77%. This shows that the ASUS TUF Gaming F15 and Lenovo can handle more tasks at once when under heavy use, making them better for multitasking.

MiniTool Partition Wizard

MiniTool Partition Wizard has a disk benchmark tool that measures the read and write performance of storage devices. In sequential read tests, the Dell Latitude E5570 performed better than other devices in general, especially when using smaller block sizes which is 4KB and 8KB. The Dell Latitude E5570 effectively managed massive file transfers with a sequential read speed of 701.889 MB/s at 32KB. The ASUS VivoBook outperformed all other systems with a sequential read speed of 1547.510 MB/s, demonstrating exceptional performance at 32KB. With a read speed of 856.863 MB/s, the Lenovo 20JTS0HT00 demonstrated strong performance, while the ASUS TUF Gaming F15 consistently delivered results with 758.438 MB/s at 32KB, without being the fastest in any particular test.

Next, for sequential write performance the Dell Latitude E5570 once again showed good performance in sequential write testing, with a write speed of 1196.51MB/s at 32KB. Despite being the fastest in sequential writing tests, the ASUS VivoBook's write speed was slightly slower, at 1105.870 MB/s. The ASUS TUF Gaming F15 achieved steady results, with a write speed of 854.405 MB/s at 32KB and 931.113 MB/s at 16 KB. At a write speed of 564.051 MB/s, the Lenovo 20JTS0HT00 displayed adequate but not exceptional performance.

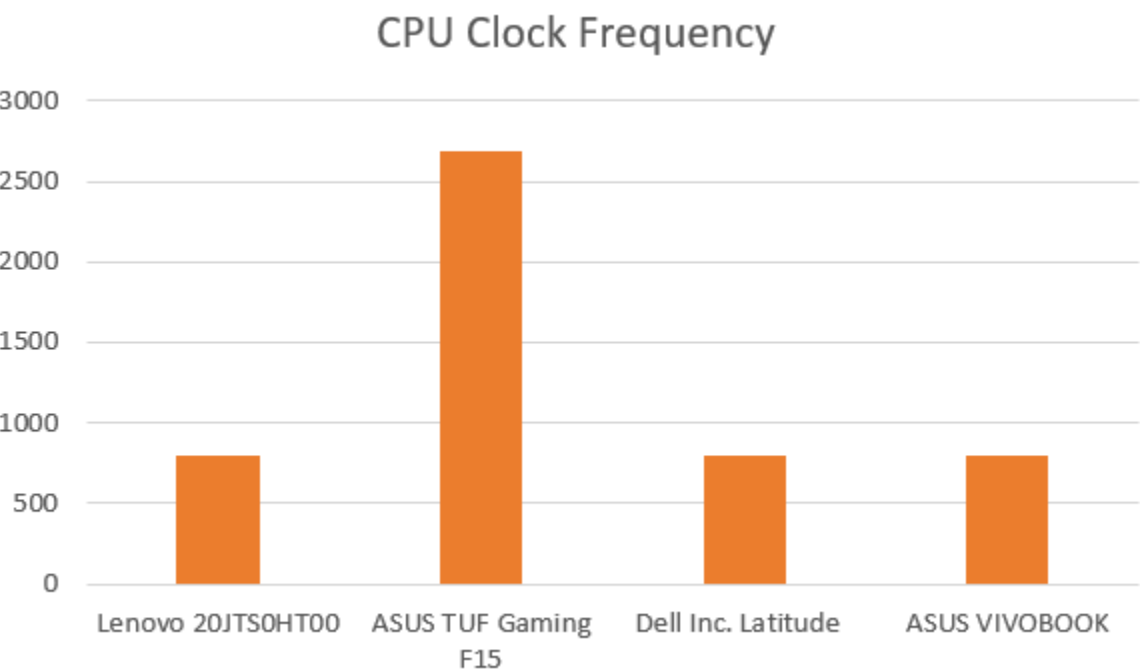
In a random reading test, the ASUS TUF Gaming F15 performed well, with a random read speed of 648.068 MB/s at 32KB. Next, the Lenovo 20JTS0HT00 delivered a good performance,

with a random read speed of 672.36 MB/s at 32KB. The Dell Latitude E5570 performed best in random read tests, especially at 32KB, with a speed of 832.38 MB/s. However, the ASUS VivoBook outperformed all other systems in random read performance at 32KB, recording in at 1095.710 MB/s. The Dell Latitude E5570 consistently beat results in smaller block sizes, such as 4KB, 8KB, and 16KB, with speeds of 650.268 MB/s, 878.987 MB/s, and 926.329 MB/s.

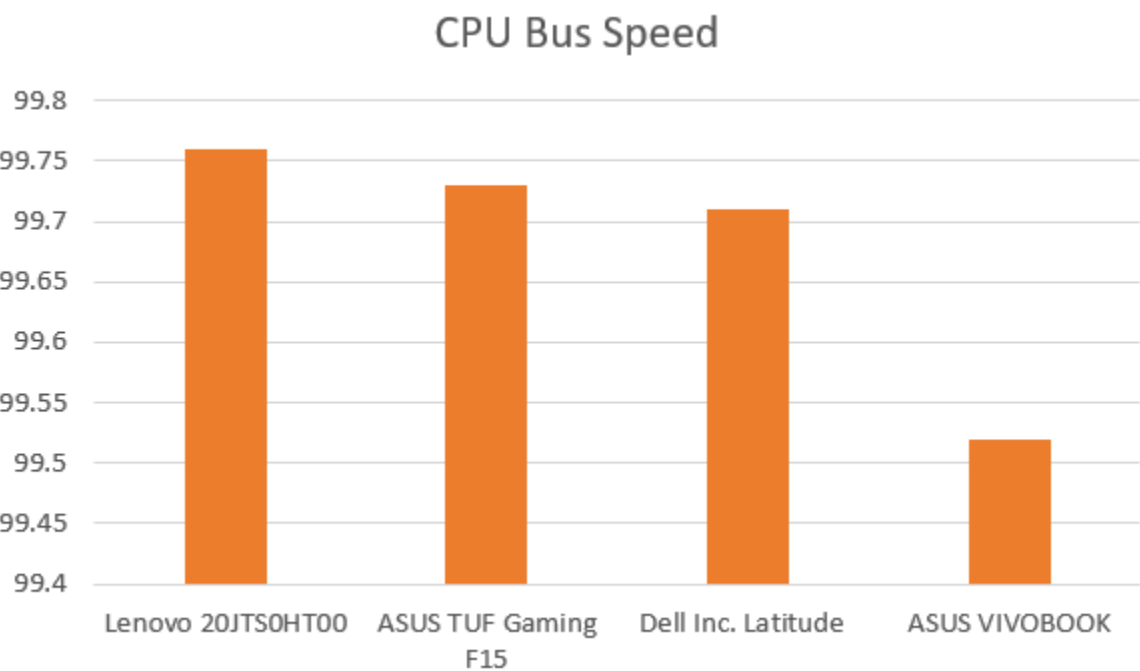
Lastly, in a random writing test, the Dell Latitude E5570 again performed well, with a speed of 1061.21 MB/s at 32KB. The ASUS TUF Gaming F15 performed well in random write operations at 32KB, with a speed of 952.713 MB/s, making it a good option in scenarios needing frequent random access writes. The Lenovo 20JTS0HT00 performed well with a random write speed of 587.691 MB/s, while the ASUS VivoBook came near to the top performers with a speed of 1047.040 MB/s at 32KB. At lower block sizes, the ASUS TUF Gaming F15 led at 4KB and 8KB with rates of 849.653 MB/s and 1001.59 MB/s, respectively, while the Dell Latitude E5570 had top speeds at 16KB and 32KB with 935.474 MB/s and 1061.21 MB/s, respectively.

Graphs

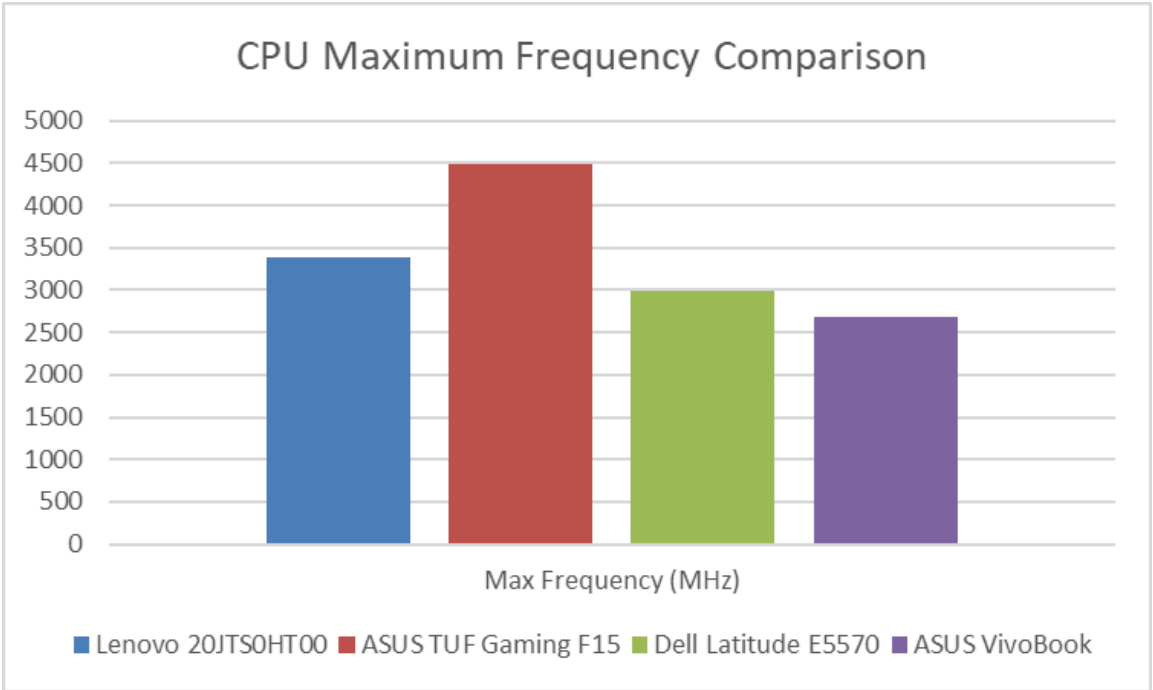
1. CPU Clock Frequency



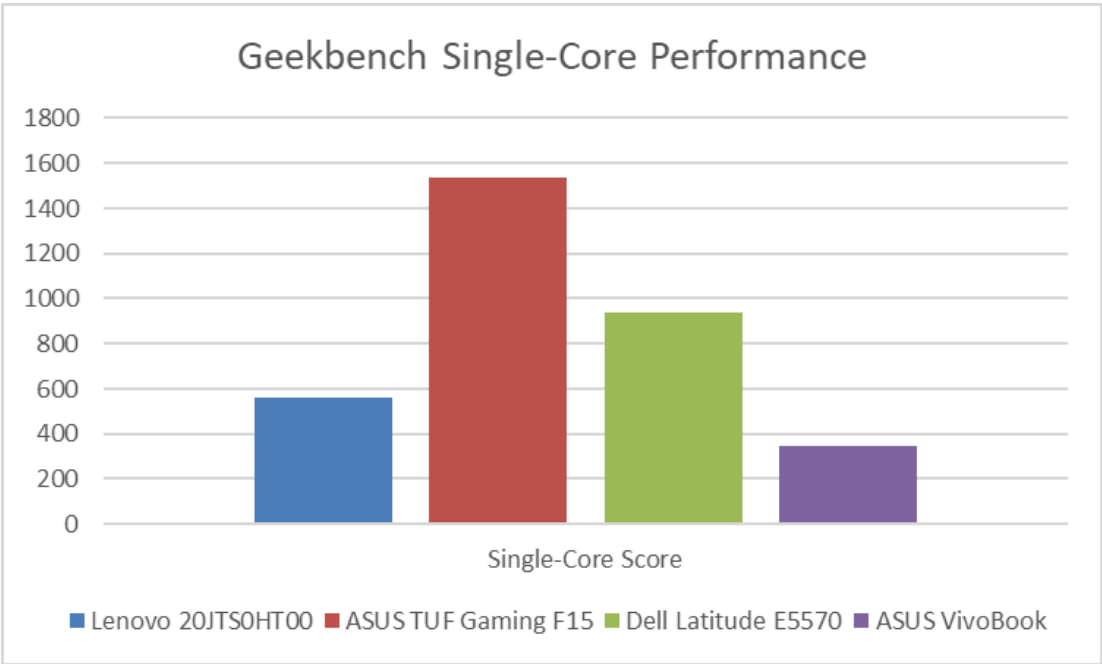
2. CPU Bus Speed



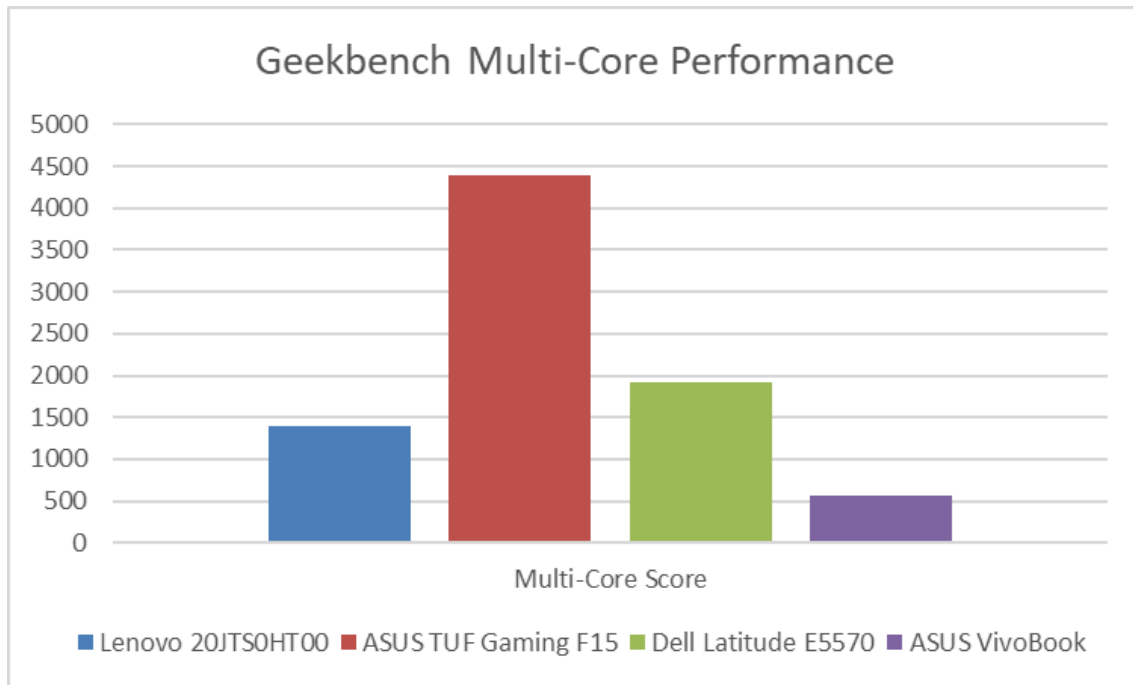
3. Geekbench CPU Maximum Frequency Comparison



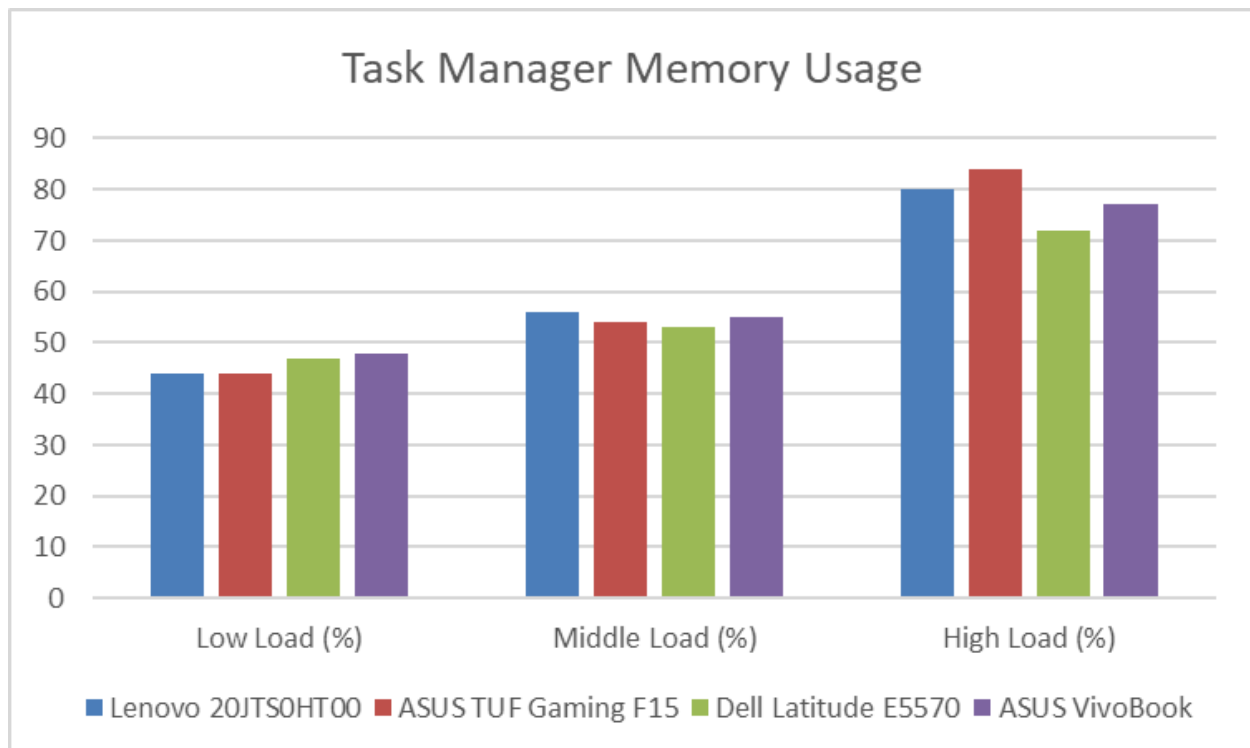
4. Geekbench Single-Core Performance



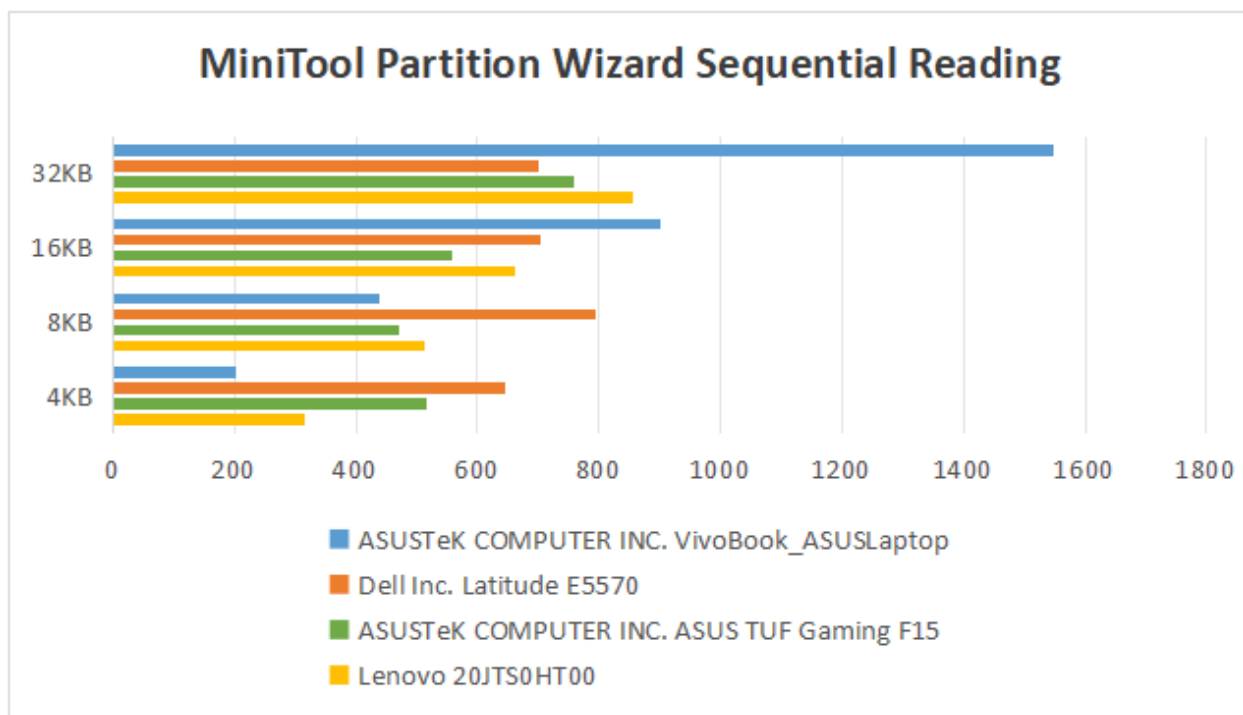
5. Geekbench Multi-Core Performance



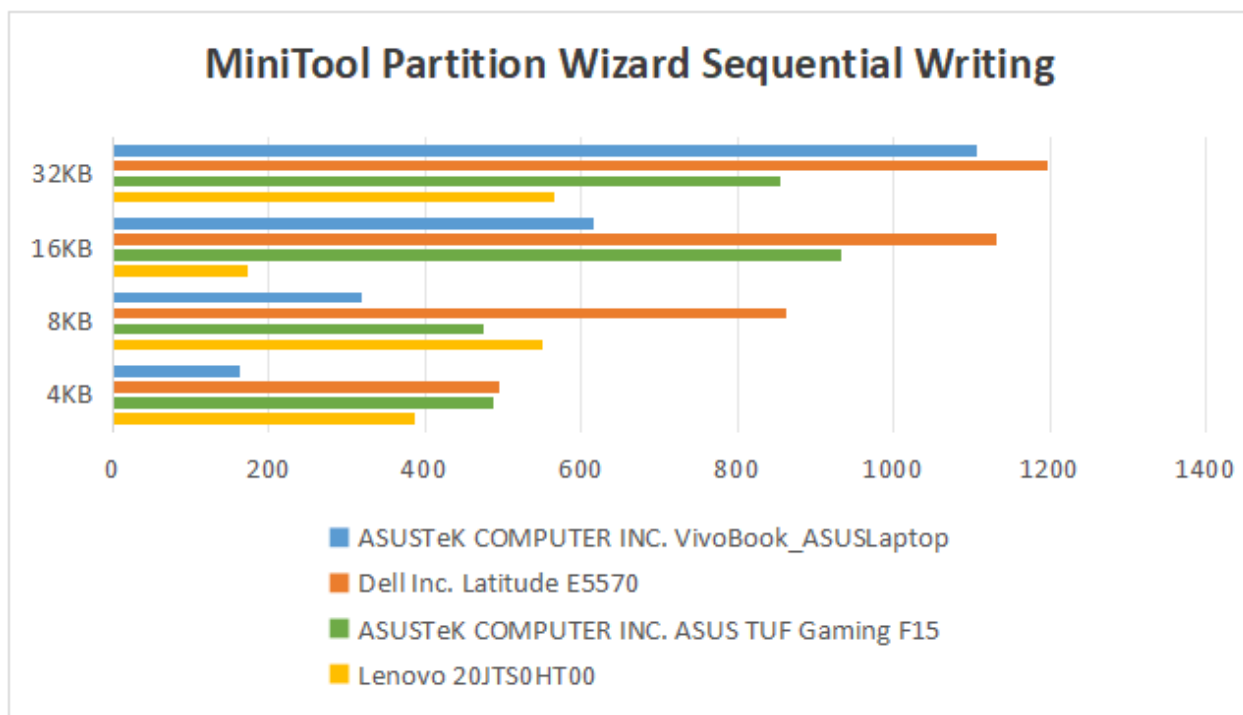
6. Task Manager Memory Usage



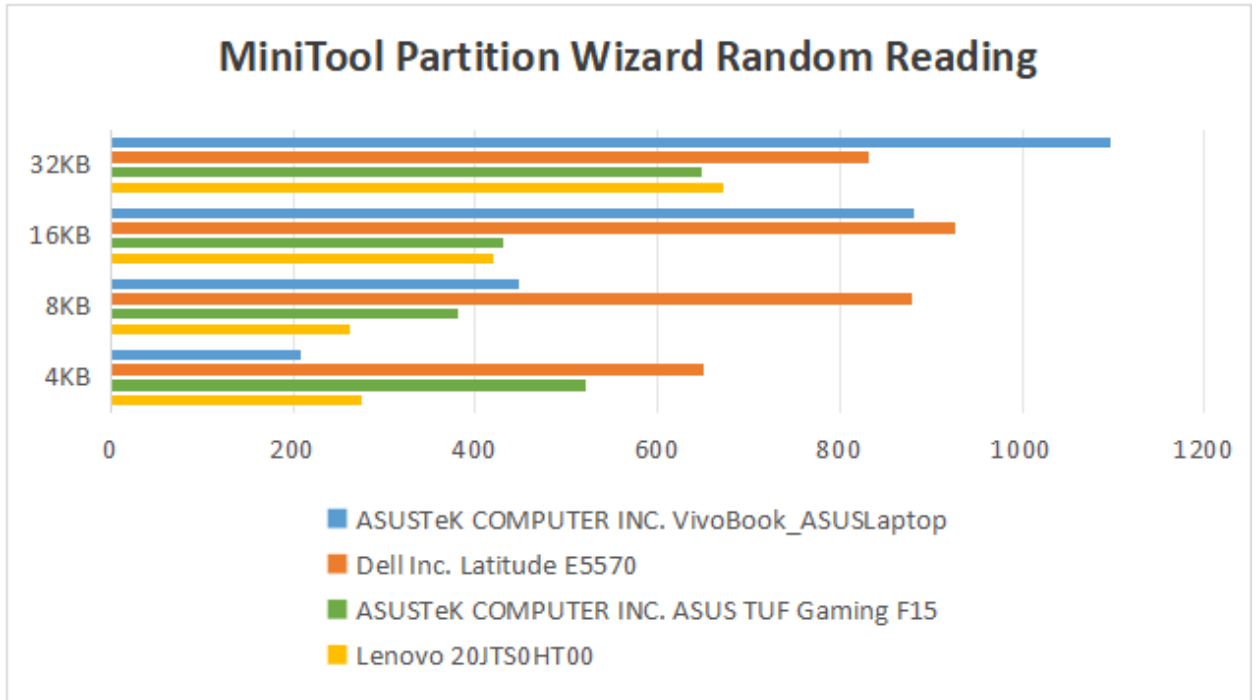
MiniTool Partition Wizard



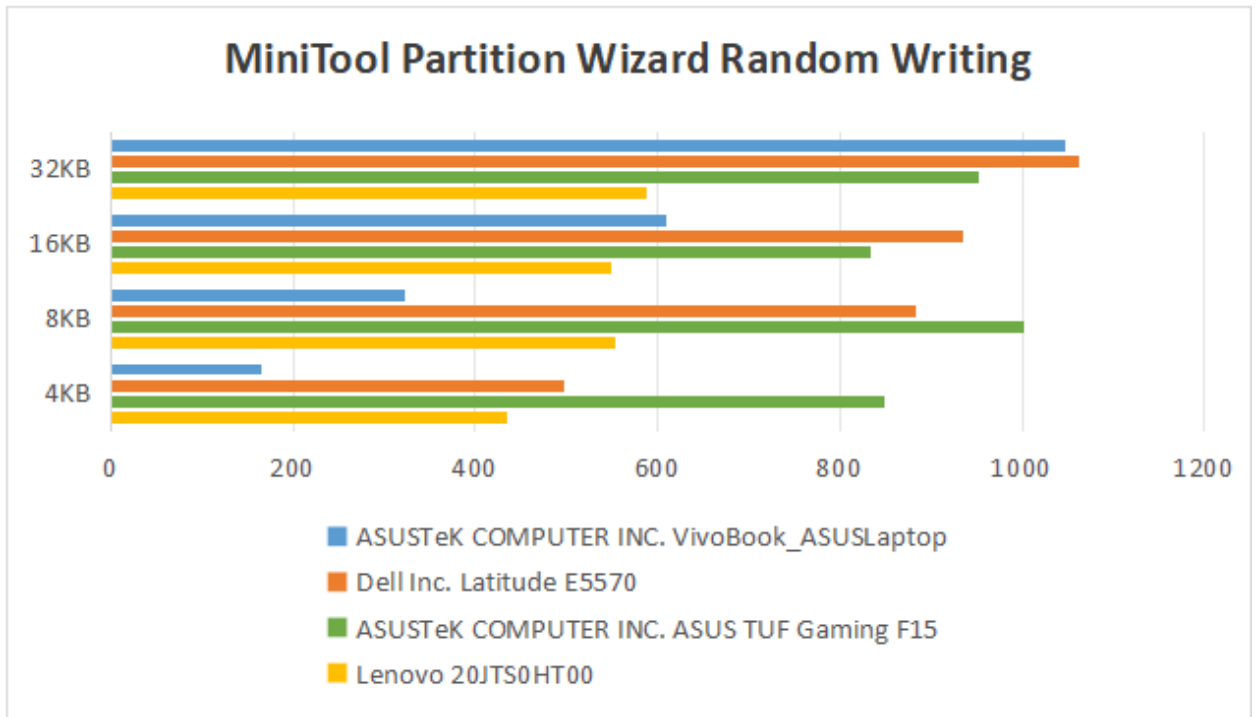
5.1 MiniTool Partition Wizard Sequential Reading



5.2 MiniTool Partition Wizard Sequential Writing



5.3 MiniTool Partition Wizard Random Reading



5.4 MiniTool Partition Wizard Random Writing

5.0 Conclusion and reflection

The benchmarking results from our study show clear differences in the performance of the evaluated CPUs. Each CPU demonstrated its own strengths across various metrics, such as clock speed, sequential reading, and random reading and writing tests. These differences highlight the importance of recognizing specific CPU capabilities to meet diverse user needs. Our findings emphasize that detailed benchmarking is crucial for informing hardware selection, ensuring users can choose the right CPU for their specific requirements and achieve optimal system performance. Understanding the nuances of CPU performance through practical, hands-on testing offers valuable insights that go beyond specifications alone, leading to more efficient and satisfying computing experiences for various tasks, from high-performance applications to routine activities.

This project underscored the critical role of CPU benchmarking in informing hardware purchase decisions. By using tools like CPU-Z, Geekbench, and MiniTool Partition Wizard, we were able to obtain detailed insights into each CPU's capabilities under various conditions. The benchmarking process was a wonderful learning experience, validating computer architecture fundamentals as well as actual implementations of these technologies. Furthermore, it demonstrated performance inequalities that are not always visible through specifications alone, underlining the importance of empirical testing.

References

CPU-Z | *Softwares* | *CPUID*. (2024, May 7). Cpuid.com.
https://cpuid.com/softwares/cpu-z.html#google_vignette

Partition Wizard. (2019). Partitionwizard.com.
<https://www.partitionwizard.com/free-partition-manager.html>

Techquickie. (2014, November 16). *Benchmarks as Fast As Possible*. YouTube.
<https://www.youtube.com/watch?v=cHtNJngtS1I>

Tweak Library. (2019, August 14). *Best CPU Benchmark Software For Windows*. YouTube. <https://www.youtube.com/watch?v=fgWGtWsDXmw>

Appendices (Tasks distribution, group pictures)

Group Members	Tasks
Dayang Farah Farzana binti Abang Idham	● Benchmarking result Task Manager ● Reflection
Farra Nurzahin bt Zaharil Anuar	● Benchmarking result and analysis CPU-Z ● Conclusion
Nurul Asyikin binti Khairul Anuar	● Benchmarking result and analysis Minitool Partition Wizard ● Flowchart
Safiya Nursyahadah binti Masnoor	● Introduction ● Benchmarking result and analysis GeekBench

